

CHAPTER 1

INTRODUCTION TO PHYSICS

1.1 WHAT IS PHYSICS ?

The nature around us is colourful and diverse. It contains phenomena of large varieties. The winds, the sands, the waters, the planets, the rainbow, heating of objects on rubbing, the function of a human body, the energy coming from the sun and the nucleus there are a large number of objects and events taking place around us.

Physics is the study of nature and its laws. We expect that all these different events in nature take place according to some basic laws and *revealing these laws of nature from the observed events* is physics. For example, the orbiting of the moon around the earth, falling of an apple from a tree and tides in a sea on a full moon night can all be explained if we know the Newton's law of gravitation and Newton's laws of motion. Physics is concerned with the basic rules which are applicable to all domains of life. Understanding of physics, therefore, leads to applications in many fields including bio and medical sciences.

The great physicist Dr R. P. Feynman has given a wonderful description of what is "understanding the nature". Suppose we do not know the rules of chess but are allowed to watch the moves of the players. If we watch the game for a long time, we may make out some of the rules. With the knowledge of these rules we may try to understand why a player played a particular move. However, this may be a very difficult task. Even if we know all the rules of chess, it is not so simple to understand all the complications of a game in a given situation and predict the correct move. Knowing the basic rules is, however, the minimum requirement if any progress is to be made.

One may guess at a wrong rule by partially watching the game. The experienced player may make use of a rule for the first time and the observer of the game may get surprised. Because of the new move some of the rules guessed at may prove to be wrong and the observer will frame new rules.

Physics goes the same way. The nature around us is like a big chess game played by Nature. The events in the nature are like the moves of the great game. We are allowed to watch the events of nature and guess at the basic rules according to which the events take place. We may come across new events which do not follow the rules guessed earlier and we may have to declare the old rules inapplicable or wrong and discover new rules.

Since physics is the study of nature, it is real. No one has been given the authority to frame the rules of physics. We only *discover* the rules that are operating in nature. Aryabhat, Newton, Einstein or Feynman are great physicists because from the observations available at that time, they could guess and frame the laws of physics which explained these observations in a convincing way. But there can be a new phenomenon any day and if the rules discovered by the great scientists are not able to explain this phenomenon, no one will hesitate to change these rules.

1.2 PHYSICS AND MATHEMATICS

The description of nature becomes easy if we have the freedom to use mathematics. To say that the gravitational force between two masses is proportional to the product of the masses and is inversely proportional to the square of the distance apart, is more difficult than to write

$$F \propto \frac{m_1 m_2}{r^2} \quad \dots (1.1)$$

Further, the techniques of mathematics such as algebra, trigonometry and calculus can be used to make predictions from the basic equations. Thus, if we know the basic rule (1.1) about the force between two particles, we can use the technique of integral calculus to find what will be the force exerted by a uniform rod on a particle placed on its perpendicular bisector.

Thus, mathematics is the language of physics. Without knowledge of mathematics it would be much more difficult to discover, understand and explain the

laws of nature. The importance of mathematics in today's world cannot be disputed. However, mathematics itself is not physics. We use a language to express our ideas. But the idea that we want to express has the main attention. If we are poor at grammar and vocabulary, it would be difficult for us to communicate our feelings but while doing so our basic interest is in the feeling that we want to express. It is nice to board a deluxe coach to go from Delhi to Agra, but the sweet memories of the deluxe coach and the video film shown on way are next to the prime goal of reaching Agra. "To understand nature" is physics, and mathematics is the deluxe coach to take us there comfortably. This relationship of physics and mathematics must be clearly understood and kept in mind while doing a physics course.

1.3 UNITS

Physics describes the laws of nature. This description is quantitative and involves measurement and comparison of physical quantities. To measure a physical quantity we need some standard unit of that quantity. An elephant is heavier than a goat but exactly how many times? This question can be easily answered if we have chosen a standard mass calling it a *unit mass*. If the elephant is 200 times the unit mass and the goat is 20 times we know that the elephant is 10 times heavier than the goat. If I have the knowledge of the unit length and some one says that Gandhi Maidan is 5 times the unit length from here, I will have the idea whether I should walk down to Gandhi Maidan or I should ride a rickshaw or I should go by a bus. Thus, the physical quantities are quantitatively expressed in terms of a unit of that quantity. The measurement of the quantity is mentioned in two parts, the first part gives how many times of the standard unit and the second part gives the name of the unit. Thus, suppose I have to study for 2 hours. The numeric part 2 says that it is 2 *times* of the unit of time and the second part *hour* says that the unit chosen here is an hour.

Who Decides the Units ?

How is a standard unit chosen for a physical quantity? The first thing is that it should have international acceptance. Otherwise, everyone will choose his or her own unit for the quantity and it will be difficult to communicate freely among the persons distributed over the world. A body named *Conférence Générale des Poids et Mesures* or CGPM also known as *General Conference on Weight and Measures* in English has been given the authority to decide the units by international agreement. It holds its meetings

and any changes in standard units are communicated through the publications of the Conference.

Fundamental and Derived Quantities

There are a large number of physical quantities which are measured and every quantity needs a definition of unit. However, not all the quantities are independent of each other. As a simple example, if a unit of length is defined, a unit of area is automatically obtained. If we make a square with its length equal to its breadth equal to the unit length, its area can be called the unit area. All areas can then be compared to this standard unit of area. Similarly, if a unit of length and a unit of time interval are defined, a unit of speed is automatically obtained. If a particle covers a unit length in unit time interval, we say that it has a unit speed. We can define a set of *fundamental quantities* as follows :

- (a) the fundamental quantities should be independent of each other, and
- (b) all other quantities may be expressed in terms of the fundamental quantities.

It turns out that the number of fundamental quantities is only seven. All the rest may be derived from these quantities by multiplication and division. Many different choices can be made for the fundamental quantities. For example, one can take speed and time as fundamental quantities. Length is then a derived quantity. If something travels at unit speed, the distance it covers in unit time interval will be called a unit distance. One may also take length and time interval as the fundamental quantities and then speed will be a derived quantity. Several systems are in use over the world and in each system the fundamental quantities are selected in a particular way. The units defined for the fundamental quantities are called *fundamental units* and those obtained for the derived quantities are called the *derived units*.

Fundamental quantities are also called base quantities.

SI Units

In 1971 CGPM held its meeting and decided a system of units which is known as the *International System of Units*. It is abbreviated as SI from the French name *Le Système International d'Unités*. This system is widely used throughout the world.

Table (1.1) gives the fundamental quantities and their units in SI.

Table 1.1 : Fundamental or Base Quantities

<u>Quantity</u>	<u>Name of the Unit</u>	<u>Symbol</u>
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric Current	ampere	A
Thermodynamic Temperature	kelvin	K
Amount of Substance	mole	mol
Luminous Intensity	candela	cd

Besides the seven fundamental units two supplementary units are defined. They are for plane angle and solid angle. The unit for plane angle is *radian* with the symbol *rad* and the unit for the solid angle is *steradian* with the symbol *sr*.

SI Prefixes

The magnitudes of physical quantities vary over a wide range. We talk of separation between two protons inside a nucleus which is about 10^{-15} m and the distance of a quasar from the earth which is about 10^{26} m. The mass of an electron is 9.1×10^{-31} kg and that of our galaxy is about 2.2×10^{41} kg. The CGPM recommended standard prefixes for certain powers of 10. Table (1.2) shows these prefixes.

Table 1.2 : SI prefixes

<u>Power of 10</u>	<u>Prefix</u>	<u>Symbol</u>
18	exa	E
15	peta	P
12	tera	T
9	giga	G
6	mega	M
3	kilo	k
2	hecto	h
1	deka	da
- 1	deci	d
- 2	centi	c
- 3	milli	m
- 6	micro	μ
- 9	nano	n
- 12	pico	p
- 15	femto	f
- 18	atto	a

1.4 DEFINITIONS OF BASE UNITS

Any standard unit should have the following two properties :

(a) *Invariability* : The standard unit must be invariable. Thus, defining distance between the tip of the middle finger and the elbow as a unit of length is not invariable.

(b) *Availability* : The standard unit should be easily made available for comparing with other quantities.

CGPM decided in its 2018 meeting that all the SI base quantities will be defined in terms of certain universal constants and these constants will be assigned fixed numerical values by definition. In this case both the criteria of invariability and availability are automatically satisfied. The new definitions became operative since 20th May 2019. We give below the definitions of these quantities. The fixed values given to the universal constants will appear in the definitions only. The definitions carry certain physical quantities and concepts that are beyond the scope of this book but you need not worry about it.

Second

1 second is the time that makes the unperturbed ground state hyperfine transition frequency $\Delta\nu_{\text{Cs}}$ to be 9192631770 when expressed in the unit Hz which is equal to s^{-1} .

Metre

1 metre is the length that makes the speed of light in vacuum to be 299792458 when expressed in the unit $\text{m}\cdot\text{s}^{-1}$, where the second is defined in terms of the caesium frequency $\Delta\nu_{\text{Cs}}$.

Kilogram

1 kilogram is the mass that makes the Planck's constant h to be $6.62607015 \times 10^{-34}$ when expressed in the unit $\text{J}\cdot\text{s}$ which is equal to $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-1}$, where the metre and the second are defined in terms of c and $\Delta\nu_{\text{Cs}}$.

Ampere

1 ampere is the current which makes the elementary charge e to be $1.602176634 \times 10^{-19}$ when expressed in the unit C which is equal to A·s, where the second is defined in terms of $\Delta\nu_{\text{Cs}}$.

Kelvin

1 kelvin is the temperature that makes the Boltzmann constant to be 1.380649×10^{-23} when expressed in the unit $\text{J}\cdot\text{K}^{-1}$ which is equal to $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-2}\cdot\text{K}^{-1}$, where kilogram, metre and second are defined in terms of h , c and $\Delta\nu_{\text{Cs}}$.

Mole

1 mole of a substance is defined to contain exactly $6.02214076 \times 10^{23}$ elementary entities. This number is the fixed numerical value of the Avogadro constant N_A when expressed in the unit mol^{-1} and is called Avogadro number.

Candela

The candela is the SI unit of luminous intensity. 1 candela is the luminous intensity that makes the luminous efficacy of monochromatic radiation of frequency 540×10^{12} Hz, K_{cd} to be 683 when expressed in the unit $\text{lm} \cdot \text{W}^{-1}$ which is equal to $\text{cd} \cdot \text{sr} \cdot \text{kg}^{-1} \cdot \text{m}^2 \cdot \text{s}^3$, where kilogram, metre and second are defined in terms of h , c and $\Delta\nu_{\text{Cs}}$.

1.5 DIMENSION

All the physical quantities of interest can be derived from the base quantities. When a quantity is expressed in terms of the base quantities, it is written as a product of different powers of the base quantities. The exponent of a base quantity that enters into the expression, is called the *dimension of the quantity in that base*. To make it clear, consider the physical quantity force. As we shall learn later, force is equal to mass times acceleration. Acceleration is change in velocity divided by time interval. Velocity is length divided by time interval. Thus,

$$\begin{aligned} \text{force} &= \text{mass} \times \text{acceleration} \\ &= \text{mass} \times \frac{\text{velocity}}{\text{time}} \\ &= \text{mass} \times \frac{\text{length/time}}{\text{time}} \\ &= \text{mass} \times \text{length} \times (\text{time})^{-2}. \quad \dots (1.2) \end{aligned}$$

Thus, the dimensions of force are 1 in mass, 1 in length and -2 in time. The dimensions in all other base quantities are zero. Note that in this type of calculation the magnitudes are not considered. It is equality of the type of quantity that enters. Thus, change in velocity, initial velocity, average velocity, final velocity all are equivalent in this discussion, each one is length/time.

For convenience the base quantities are represented by one letter symbols. Generally, mass is denoted by M , length by L , time by T and electric current by I . The thermodynamic temperature, the amount of substance and the luminous intensity are denoted by the symbols of their units K , mol and cd respectively. The physical quantity that is expressed in terms of the base quantities is enclosed in square

brackets to remind that the equation is among the dimensions and not among the magnitudes. Thus equation (1.2) may be written as $[\text{force}] = \text{MLT}^{-2}$.

Such an expression for a physical quantity in terms of the base quantities is called the *dimensional formula*. Thus, the dimensional formula of force is MLT^{-2} . The two versions given below are equivalent and are used interchangeably.

(a) The dimensional formula of force is MLT^{-2} .

(b) The dimensions of force are 1 in mass, 1 in length and -2 in time.

Example 1.1

Calculate the dimensional formula of energy from the equation $E = \frac{1}{2}mv^2$.

Solution : Dimensionally, $E = \text{mass} \times (\text{velocity})^2$, since $\frac{1}{2}$ is a number and has no dimension.

$$\therefore [E] = M \times \left(\frac{L}{T}\right)^2 = \text{ML}^2 \text{T}^{-2}.$$

1.6 USES OF DIMENSION**A. Homogeneity of Dimensions in an Equation**

An equation contains several terms which are separated from each other by the symbols of equality, plus or minus. The dimensions of all the terms in an equation must be identical. This is another way of saying that one can add or subtract similar physical quantities. Thus, a velocity cannot be added to a force or an electric current cannot be subtracted from the thermodynamic temperature. This simple principle is called the *principle of homogeneity of dimensions* in an equation and is an extremely useful method to check whether an equation may be correct or not. If the dimensions of all the terms are not same, the equation must be wrong. Let us check the equation

$$x = ut + \frac{1}{2}at^2$$

for the dimensional homogeneity. Here x is the distance travelled by a particle in time t which starts at a speed u and has an acceleration a along the direction of motion.

$$[x] = L$$

$$[ut] = \text{velocity} \times \text{time} = \frac{\text{length}}{\text{time}} \times \text{time} = L$$

$$\begin{aligned} \left[\frac{1}{2}at^2\right] &= [at^2] = \text{acceleration} \times (\text{time})^2 \\ &= \frac{\text{velocity}}{\text{time}} \times (\text{time})^2 = \frac{\text{length/time}}{\text{time}} \times (\text{time})^2 = L \end{aligned}$$

Thus, the equation is correct as far as the dimensions are concerned.

Limitation of the Method

Note that the dimension of $\frac{1}{2}at^2$ is same as that of at^2 . Pure numbers are dimensionless. Dimension does not depend on the magnitude. Due to this reason the equation $x = ut + at^2$ is also dimensionally correct. Thus, a dimensionally correct equation need not be actually correct but a dimensionally wrong equation must be wrong.

Example 1.2

Test dimensionally if the formula $t = 2\pi\sqrt{\frac{m}{F/x}}$ may be correct, where t is time period, m is mass, F is force and x is distance.

Solution : The dimension of force is MLT^{-2} . Thus, the dimension of the right-hand side is

$$\sqrt{\frac{M}{MLT^{-2}/L}} = \sqrt{\frac{1}{T^{-2}}} = T.$$

The left-hand side is time period and hence the dimension is T . The dimensions of both sides are equal and hence the formula may be correct.

B. Conversion of Units

When we choose to work with a different set of units for the base quantities, the units of all the derived quantities must be changed. Dimensions can be useful in finding the conversion factor for the unit of a derived physical quantity from one system to other. Consider an example. When SI units are used, the unit of pressure is 1 pascal. Suppose we choose 1 cm as the unit of length, 1 g as the unit of mass and 1 s as the unit of time (this system is still in wide use and is called CGS system). The unit of pressure will be different in this system. Let us call it for the time-being 1 CGS pressure. Now, how many CGS pressure is equal to 1 pascal ?

Let us first write the dimensional formula of pressure.

We have
$$P = \frac{F}{A}.$$

Thus,
$$[P] = \frac{[F]}{[A]} = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$$

so,
$$1 \text{ pascal} = (1 \text{ kg}) (1 \text{ m})^{-1} (1 \text{ s})^{-2}$$

and
$$1 \text{ CGS pressure} = (1 \text{ g}) (1 \text{ cm})^{-1} (1 \text{ s})^{-2}$$

Thus,
$$\frac{1 \text{ pascal}}{1 \text{ CGS pressure}} = \left(\frac{1 \text{ kg}}{1 \text{ g}}\right) \left(\frac{1 \text{ m}}{1 \text{ cm}}\right)^{-1} \left(\frac{1 \text{ s}}{1 \text{ s}}\right)^{-2}$$

$$= (10^3) (10^2)^{-1} = 10$$

or,
$$1 \text{ pascal} = 10 \text{ CGS pressure}.$$

Thus, knowing the conversion factors for the base quantities, one can work out the conversion factor for any derived quantity if the dimensional formula of the derived quantity is known.

C. Deducing Relation among the Physical Quantities

Sometimes dimensions can be used to deduce a relation between the physical quantities. If one knows the quantities on which a particular physical quantity depends and if one guesses that this dependence is of product type, method of dimension may be helpful in the derivation of the relation. Taking an example, suppose we have to derive the expression for the time period of a simple pendulum. The simple pendulum has a bob, attached to a string, which oscillates under the action of the force of gravity. Thus, the time period may depend on the length of the string, the mass of the bob and the acceleration due to gravity. We assume that the dependence of time period on these quantities is of product type, that is,

$$t = kl^a m^b g^c \quad \dots (1.3)$$

where k is a dimensionless constant and a , b and c are exponents which we want to evaluate. Taking the dimensions of both sides,

$$T = L^a M^b (LT^{-2})^c = L^{a+c} M^b T^{-2c}.$$

Since the dimensions on both sides must be identical, we have

$$a + c = 0$$

$$b = 0$$

$$\text{and} \quad -2c = 1$$

$$\text{giving } a = \frac{1}{2}, \quad b = 0 \quad \text{and} \quad c = -\frac{1}{2}.$$

Putting these values in equation (1.3)

$$t = k \sqrt{\frac{l}{g}}. \quad \dots (1.4)$$

Thus, by dimensional analysis we can deduce that the time period of a simple pendulum is independent of its mass, is proportional to the square root of the length of the pendulum and is inversely proportional to the square root of the acceleration due to gravity at the place of observation.

Limitations of the Dimensional Method

Although dimensional analysis is very useful in deducing certain relations, it cannot lead us too far. First of all we have to know the quantities on which a particular physical quantity depends. Even then the method works only if the dependence is of the product type. For example, the distance travelled by a uniformly accelerated particle depends on the initial velocity u , the acceleration a and the time t . But the method of dimensions cannot lead us to the correct expression for x because the expression is not of

product type. It is equal to the sum of two terms as

$$x = ut + \frac{1}{2}at^2.$$

Secondly, the numerical constants having no dimensions cannot be deduced by the method of dimensions. In the example of time period of a simple pendulum, an unknown constant k remains in equation (1.4). One has to know from somewhere else that this constant is 2π .

Thirdly, the method works only if there are as many equations available as there are unknowns. In mechanical quantities, only three base quantities length, mass and time enter. So, dimensions of these three may be equated in the guessed relation giving at most three equations in the exponents. If a particular quantity (in mechanics) depends on more than three quantities we shall have more unknowns and less equations. The exponents cannot be determined uniquely in such a case. Similar constraints are present for electrical or other nonmechanical quantities.

1.7 ORDER OF MAGNITUDE

In physics, we come across quantities which vary over a wide range. We talk of the size of a mountain and the size of the tip of a pin. We talk of the mass of our galaxy and the mass of a hydrogen atom. We talk of the age of the universe and the time taken by an electron to complete a circle around the proton in a hydrogen atom. It becomes quite difficult to get a feel of largeness or smallness of such quantities. To express such widely varying numbers, one uses the *powers of ten* method.

In this method, each number is expressed as $a \times 10^b$ where $1 \leq a < 10$ and b is a positive or negative integer. Thus the diameter of the sun is expressed as 1.39×10^9 m and the diameter of a hydrogen atom as 1.06×10^{-10} m. To get an approximate idea of the number, one may round the number a to 1 if it is less than or equal to 5 and to 10 if it is greater than 5. The number can then be expressed approximately as 10^b . We then get the *order of magnitude* of that number. Thus, the diameter of the sun is *of the order of* 10^9 m and that of a hydrogen atom is *of the order of* 10^{-10} m. More precisely, the exponent of 10 in such a representation is called the order of magnitude of that quantity. Thus, the diameter of the sun is 19 *orders of magnitude larger* than the diameter of a hydrogen atom. This is because the order of magnitude of 10^9 is 9 and of 10^{-10} is -10 . The difference is $9 - (-10) = 19$.

To quickly get an approximate value of a quantity in a given physical situation, one can make an *order*

of magnitude calculation. In this all numbers are approximated to 10^b form and the calculation is made.

Let us estimate the number of persons that may sit in a circular field of radius 800 m. The area of the field is

$$A = \pi r^2 = 3.14 \times (800 \text{ m})^2 \approx 10^6 \text{ m}^2.$$

The average area one person occupies in sitting $\approx 50 \text{ cm} \times 50 \text{ cm} = 0.25 \text{ m}^2 = 2.5 \times 10^{-1} \text{ m}^2 \approx 10^{-1} \text{ m}^2$.

The number of persons who can sit in the field is

$$N \approx \frac{10^6 \text{ m}^2}{10^{-1} \text{ m}^2} = 10^7.$$

Thus of the order of 10^7 persons may sit in the field.

1.8 THE STRUCTURE OF WORLD

Man has always been interested to find how the world is structured. Long long ago scientists suggested that the world is made up of certain indivisible small particles. The number of particles in the world is large but the varieties of particles are not many. Old Indian philosopher Kanadi derives his name from this proposition (In Sanskrit or Hindi *Kana* means a small particle). After extensive experimental work people arrived at the conclusion that the world is made up of just three types of ultimate particles, the proton, the neutron and the electron. All objects which we have around us, are aggregation of atoms and molecules. The molecules are composed of atoms and the atoms have at their heart a nucleus containing protons and neutrons. Electrons move around this nucleus in special arrangements. It is the number of protons, neutrons and electrons in an atom that decides all the properties and behaviour of a material. Large number of atoms combine to form an object of moderate or large size. However, the laws that we generally deduce for these macroscopic objects are not always applicable to atoms, molecules, nuclei or the elementary particles. These laws known as *classical physics* deal with large size objects only. When we say a particle in classical physics we mean an object which is small as compared to other moderate or large size objects and for which the classical physics is valid. It may still contain millions and millions of atoms in it. Thus, a particle of dust dealt in classical physics may contain about 10^{18} atoms.

Twentieth century experiments have revealed another aspect of the construction of world. There are perhaps no ultimate indivisible particles. Hundreds of elementary particles have been discovered and there are free transformations from one such particle to the other. Nature is seen to be a well-connected entity.

CHAPTER 3

REST AND MOTION : KINEMATICS

3.1 REST AND MOTION

When do we say that a body is at rest and when do we say that it is in motion? You may say that if a body does not change its position as time passes it is at rest. If a body changes its position with time, it is said to be moving. But when do we say that it is not changing its position? A book placed on the table remains on the table and we say that the book is at rest. However, if we station ourselves on the moon (the Appollo missions have made it possible), the whole earth is found to be changing its position and so the room, the table and the book are all continuously changing their positions. The book is at rest if it is viewed from the room, it is moving if it is viewed from the moon.

Motion is a combined property of the object under study and the observer. There is no meaning of rest or motion without the viewer. Nothing is in absolute rest or in absolute motion. The moon is moving with respect to the book and the book moves with respect to the moon. Take another example. A robber enters a train moving at great speed with respect to the ground, brings out his pistol and says “Don’t move, stand still”. The passengers stand still. The passengers are at rest with respect to the robber but are moving with respect to the rail track.

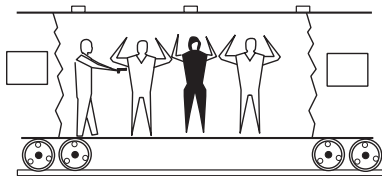


Figure 3.1

To locate the position of a particle we need a *frame of reference*. A convenient way to fix up the frame of reference is to choose three mutually perpendicular axes and name them X-Y-Z axes. The coordinates, (x, y, z) of the particle then specify the position of the

particle with respect to that frame. Add a clock into the frame of reference to measure the time. If all the three coordinates x, y and z of the particle remain unchanged as time passes, we say that the particle is at rest with respect to this frame. If any one or more coordinates change with time, we say that the body is moving with respect to this frame.

There is no rule or restriction on the choice of a frame. We can choose a frame of reference according to our convenience to describe the situation under study. Thus, when we are in a train it is convenient to choose a frame attached to our compartment. The coordinates of a suitcase placed on the upper berth do not change with time (unless the train gives a jerk) and we say that the suitcase is at rest in the train-frame. The different stations, electric poles, trees etc. change their coordinates and we say that they are moving in the train-frame. Thus, we say that “Bombay is coming” and “Pune has already passed”.

In the following sections we shall assume that the frame of reference is already chosen and we are describing the motion of the objects in the chosen frame. Sometimes the choice of the frame is clear from the context and we do not mention it. Thus, when one says the car is travelling and the rickshaw is not, it is clear that all positions are measured from a frame attached to the road.

3.2 DISTANCE AND DISPLACEMENT

Suppose a particle is at A at time t_1 and at B at time t_2 with respect to a given frame (figure 3.2).

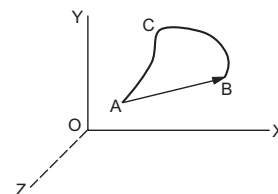


Figure 3.2

During the time interval t_1 to t_2 the particle moves along the path ACB . The length of the path ACB is called the *distance* travelled during the time interval t_1 to t_2 . If we connect the initial position A with the final position B by a straight line, we get the *displacement* of the particle. The magnitude of the displacement is the length of the straight line joining the initial and the final position. The direction is from the initial to the final position. The displacement has both the magnitude as well as the direction. Further the displacements add according to the triangle rule of vector addition. Suppose a particle kept on a table is displaced on the table and at the same time the table is also displaced in the room. The net displacement of the particle in the room is obtained by the vector sum of the two displacements. Thus, displacement is a vector quantity. In contrast the distance covered has only a magnitude and is thus, a scalar quantity.

Example 3.1

An old person moves on a semi-circular track of radius 40.0 m during a morning walk. If he starts at one end of the track and reaches at the other end, find the distance covered and the displacement of the person.

Solution : The distance covered by the person equals the length of the track. It is equal to $\pi R = \pi \times 40.0 \text{ m} = 126 \text{ m}$.

The displacement is equal to the diameter of the semi-circular track joining the two ends. It is $2R = 2 \times 40.0 \text{ m} = 80 \text{ m}$. The direction of this displacement is from the initial point to the final point.

3.3 AVERAGE SPEED AND INSTANTANEOUS SPEED

The average speed of a particle in a time interval is defined as the distance travelled by the particle divided by the time interval. If the particle travels a distance s in time t_1 to t_2 , the *average speed* is defined as

$$v_{av} = \frac{s}{t_2 - t_1} \quad \dots (3.1)$$

The average speed gives the overall “rapidity” with which the particle moves in this interval. In a one-day cricket match, the average run rate is quoted as the total runs divided by the total number of overs used to make these runs. Some of the overs may be expensive and some may be economical. Similarly, the average speed gives the total effect in the given interval. The rapidity or slowness may vary from instant to instant. When an athlete starts running, he or she runs slowly and gradually

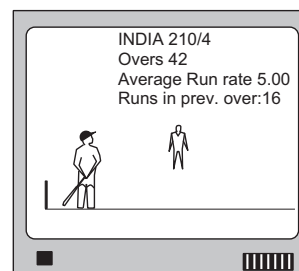


Figure 3.3

increases the rate. We define the *instantaneous speed* at a time t as follows.

Let Δs be the distance travelled in the time interval t to $t + \Delta t$. The average speed in this time interval is

$$v_{av} = \frac{\Delta s}{\Delta t}.$$

Now make Δt vanishingly small and look for the value of $\frac{\Delta s}{\Delta t}$. Remember Δs is the distance travelled in the chosen time interval Δt . As Δt approaches 0, the distance Δs also approaches zero but the ratio $\frac{\Delta s}{\Delta t}$ has a finite limit.

The instantaneous speed at a time t is defined as

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt} \quad \dots (3.2)$$

where s is the distance travelled in time t . The average speed is defined for a time interval and the instantaneous speed is defined at a particular instant. Instantaneous speed is also called “speed”.

Example 3.2

The distance travelled by a particle in time t is given by $s = (2.5 \text{ m/s}^2) t^2$. Find (a) the average speed of the particle during the time 0 to 5.0 s, and (b) the instantaneous speed at $t = 5.0 \text{ s}$.

Solution : (a) The distance travelled during time 0 to 5.0 s is

$$s = (2.5 \text{ m/s}^2) (5.0 \text{ s})^2 = 62.5 \text{ m}.$$

The average speed during this time is

$$v_{av} = \frac{62.5 \text{ m}}{5 \text{ s}} = 12.5 \text{ m/s}.$$

$$(b) \quad s = (2.5 \text{ m/s}^2) t^2$$

$$\text{or,} \quad \frac{ds}{dt} = (2.5 \text{ m/s}^2) (2t) = (5.0 \text{ m/s}^2) t.$$

At $t = 5.0 \text{ s}$ the speed is

$$v = \frac{ds}{dt} = (5.0 \text{ m/s}^2) (5.0 \text{ s}) = 25 \text{ m/s}.$$

If we plot the distance s as a function of time (figure 3.4), the speed at a time t equals the slope of

the tangent to the curve at the time t . The average speed in a time interval t to $t + \Delta t$ equals the slope of the chord AB where A and B are the points on the

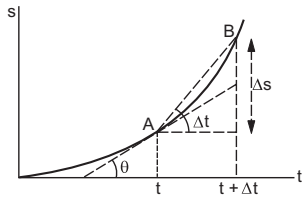


Figure 3.4

curve corresponding to the time t and $t + \Delta t$. As Δt approaches zero, the chord AB becomes the tangent at A and the average speed $\frac{\Delta s}{\Delta t}$ becomes the slope of the tangent which is $\frac{ds}{dt}$.

If the speed of the particle at time t is v , the distance ds travelled by it in the short time interval t to $t + dt$ is $v dt$. Thus, $ds = v dt$. The total distance travelled by the particle in a finite time interval t_1 to t_2 can be obtained by summing over these small distances ds as time changes from t_1 to t_2 . Thus, the distance travelled by a particle in the time interval

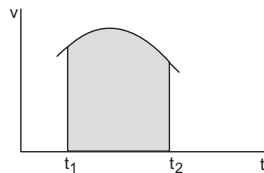


Figure 3.5

t_1 to t_2 is

$$s = \int_{t_1}^{t_2} v dt. \quad \dots (3.3)$$

If we plot a graph of the speed v versus time t , the distance travelled by the particle can be obtained by finding the area under the curve. Figure (3.5) shows such a speed-time graph. To find the distance travelled in the time interval t_1 to t_2 we draw ordinates from $t = t_1$ and $t = t_2$. The area bounded by the curve $v - t$, the X-axis and the two ordinates at $t = t_1$ and $t = t_2$ (shown shaded in the figure) gives the total distance covered.

The dimension of speed is LT^{-1} and its SI unit is metre/second abbreviated as m/s.

Example 3.3

Figure (3.6) shows the speed versus time graph for a particle. Find the distance travelled by the particle during the time $t = 0$ to $t = 3$ s.

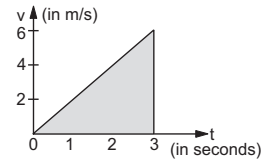


Figure 3.6

Solution : The distance travelled by the particle in the time 0 to 3 s is equal to the area shaded in the figure. This is a right angled triangle with height = 6 m/s and the base = 3 s. The area is $\frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2} \times (3 \text{ s})(6 \text{ m/s}) = 9 \text{ m}$. Thus, the particle covered a distance of 9 m during the time 0 to 3 s.

3.4 AVERAGE VELOCITY AND INSTANTANEOUS VELOCITY

The *average velocity* of a particle in a time interval t_1 to t_2 is defined as its displacement divided by the time interval. If the particle is at a point A (figure 3.7) at time $t = t_1$ and at B at time $t = t_2$, the displacement in this time interval is the vector $\vec{AB}$. The average velocity in this time interval is then,

$$\vec{v}_{av} = \frac{\vec{AB}}{t_2 - t_1}.$$

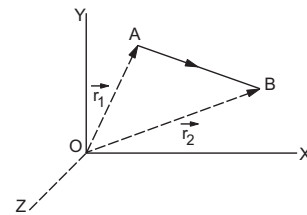


Figure 3.7

Like displacement, it is a vector quantity.

Position vector : If we join the origin to the position of the particle by a straight line and put an arrow towards the position of the particle, we get the *position vector* of the particle. Thus, the position vector of the particle shown in figure (3.7) at time $t = t_1$ is $\vec{OA}$ and that at $t = t_2$ is $\vec{OB}$. The displacement of the particle in the time interval t_1 to t_2 is

$$\vec{AB} = \vec{AO} + \vec{OB} = \vec{OB} - \vec{OA} = \vec{r}_2 - \vec{r}_1.$$

The average velocity of a particle in the time interval t_1 to t_2 can be written as

$$\vec{v}_{av} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} \quad \dots (3.4)$$

Note that only the positions of the particle at time $t = t_1$ and $t = t_2$ are used in calculating the average velocity. The positions in between t_1 and t_2 are not needed, hence the actual path taken in going from A to B is not important in calculating the average velocity.

Example 3.4

A table clock has its minute hand 4.0 cm long. Find the average velocity of the tip of the minute hand (a) between 6:00 a.m. to 6:30 a.m. and (b) between 6:00 a.m. to 6:30 p.m.

Solution : At 6:00 a.m. the tip of the minute hand is at 12 mark and at 6:30 a.m. or 6:30 p.m. it is 180° away. Thus, the straight line distance between the initial and final position of the tip is equal to the diameter of the clock.

Displacement = $2R = 2 \times 4.0 \text{ cm} = 8.0 \text{ cm}$.

The displacement is from the 12 mark to the 6 mark on the clock panel. This is also the direction of the average velocity in both cases.

(a) The time taken from 6:00 a.m. to 6:30 a.m. is 30 minutes = 1800 s. The average velocity is

$$v_{av} = \frac{\text{Displacement}}{\text{time}} = \frac{8.0 \text{ cm}}{1800 \text{ s}} = 4.4 \times 10^{-3} \text{ cm/s}.$$

(b) The time taken from 6:00 a.m. to 6:30 p.m. is 12 hours and 30 minutes = 45000 s. The average velocity is

$$v_{av} = \frac{\text{Displacement}}{\text{time}} = \frac{8.0 \text{ cm}}{45000 \text{ s}} = 1.8 \times 10^{-4} \text{ cm/s}.$$

The *instantaneous velocity* of a particle at a time t is defined as follows. Let the average velocity of the particle in a short time interval t to $t + \Delta t$ be $\vec{v}_{av}$. This average velocity can be written as

$$\vec{v}_{av} = \frac{\Delta \vec{r}}{\Delta t}$$

where $\Delta \vec{r}$ is the displacement in the time interval Δt . We now make Δt vanishingly small and find the limiting value of $\frac{\Delta \vec{r}}{\Delta t}$. This value is instantaneous velocity $\vec{v}$ of the particle at time t .

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt} \quad \dots (3.5)$$

For very small intervals the displacement $\Delta \vec{r}$ is along the line of motion of the particle. Thus, the length

Δr equals the distance Δs travelled in that interval. So the magnitude of the velocity is

$$v = \left| \frac{d\vec{r}}{dt} \right| = \frac{|d\vec{r}|}{dt} = \frac{ds}{dt} \quad \dots (3.6)$$

which is the instantaneous speed at time t . Instantaneous velocity is also called the “velocity”.

3.5 AVERAGE ACCELERATION AND INSTANTANEOUS ACCELERATION

If the velocity of a particle remains constant as time passes, we say that it is moving with uniform velocity. If the velocity changes with time, it is said to be accelerated. The acceleration is the rate of change of velocity. Velocity is a vector quantity hence a change in its magnitude or direction or both will change the velocity.

Suppose the velocity of a particle at time t_1 is $\vec{v}_1$ and at time t_2 it is $\vec{v}_2$. The change produced in time interval t_1 to t_2 is $\vec{v}_2 - \vec{v}_1$. We define the *average acceleration* $\vec{a}_{av}$ as the change in velocity divided by the time interval. Thus,

$$\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} \quad \dots (3.7)$$

Again the average acceleration depends only on the velocities at time t_1 and t_2 . How the velocity changed in between t_1 and t_2 is not important in defining the average acceleration.

Instantaneous acceleration of a particle at time t is defined as

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} \quad \dots (3.8)$$

where $\Delta \vec{v}$ is the change in velocity between the time t and $t + \Delta t$. At time t the velocity is $\vec{v}$ and at time $t + \Delta t$ it becomes $\vec{v} + \Delta \vec{v}$. $\frac{\Delta \vec{v}}{\Delta t}$ is the average acceleration

of the particle in the interval Δt . As Δt approaches zero, this average acceleration becomes the instantaneous acceleration. Instantaneous acceleration is also called “acceleration”.

The dimension of acceleration is LT^{-2} and its SI unit is metre/second² abbreviated as m/s².

3.6 MOTION IN A STRAIGHT LINE

When a particle is constrained to move on a straight line, the description becomes fairly simple. We choose the line as the X-axis and a suitable time instant as $t = 0$. Generally the origin is taken at the point where the particle is situated at $t = 0$. The position of the particle at time t is given by its coordinate x at that time. The velocity is

$$v = \frac{dx}{dt} \quad \dots (3.9)$$

$$\text{and the acceleration is } a = \frac{dv}{dt} \quad \dots (3.10)$$

$$= \frac{d}{dt} \left(\frac{dx}{dt} \right) = \frac{d^2 x}{dt^2} \quad \dots (3.11)$$

If $\frac{dx}{dt}$ is positive, the direction of the velocity is along the positive X-axis and if $\frac{dx}{dt}$ is negative, the direction, is along the negative X-axis. Similarly if $\frac{dv}{dt}$ is positive, the acceleration is along the positive X-axis and if $\frac{dv}{dt}$ is negative, the acceleration is along the negative X-axis. The magnitude of v is speed. If the velocity and the acceleration are both positive, the speed increases. If both of them are negative then also the speed increases but if they have opposite signs, the speed decreases. When the speed decreases, we say that the particle is decelerating. Deceleration is equivalent to negative acceleration. An acceleration of 2.0 m/s^2 towards east is same as a deceleration of 2.0 m/s^2 towards west.

Motion with Constant Acceleration

Suppose the acceleration of a particle is a and remains constant. Let the velocity at time 0 be u and the velocity at time t be v . Thus,

$$\frac{dv}{dt} = a, \quad \text{or,} \quad dv = a dt$$

$$\text{or,} \quad \int_u^v dv = \int_0^t a dt.$$

As time changes from 0 to t the velocity changes from u to v . So on the left hand side the summation is made over v from u to v whereas on the right hand side the summation is made on time from 0 to t . Evaluating the integrals we get,

$$[v]_u^v = a[t]_0^t$$

$$\text{or,} \quad v - u = at$$

$$\text{or,} \quad v = u + at. \quad \dots (3.12)$$

Equation (3.12) may be written as

$$\frac{dx}{dt} = u + at$$

$$\text{or,} \quad dx = (u + at)dt$$

$$\text{or,} \quad \int_0^x dx = \int_0^t (u + at)dt.$$

At $t = 0$ the particle is at $x = 0$. As time changes from 0 to t the position changes from 0 to x . So on the left hand side the summation is made on position from

0 to x whereas on the right hand side the summation is made on time from 0 to t . Evaluating the integrals, the above equation becomes

$$[x]_0^x = \int_0^t u dt + \int_0^t at dt$$

$$\text{or,} \quad x = u \int_0^t dt + a \int_0^t t dt$$

$$= u[t]_0^t + a \left[\frac{t^2}{2} \right]_0^t$$

$$\text{or,} \quad x = ut + \frac{1}{2} at^2. \quad \dots (3.13)$$

From equation (3.12),

$$v^2 = (u + at)^2$$

$$\text{or,} \quad = u^2 + 2uat + a^2 t^2$$

$$\text{or,} \quad = u^2 + 2a \left(ut + \frac{1}{2} at^2 \right)$$

$$\text{or,} \quad = u^2 + 2ax. \quad \dots (3.14)$$

The three equations (3.12) to (3.14) are collected below in table 3.1. They are very useful in solving the problems of motion in a straight line with constant acceleration.

Table 3.1

$v = u + at$ $x = ut + \frac{1}{2} at^2$ $v^2 = u^2 + 2ax$
--

Remember that x represents the position of the particle at time t and not (in general) the distance travelled by it in time 0 to t . For example, if the particle starts from the origin and goes upto $x = 4 \text{ m}$, then turns and is at $x = 2 \text{ m}$ at time t , the distance travelled is 6 m but the position is still given by $x = 2 \text{ m}$.

The quantities u , v and a may take positive or negative values depending on whether they are directed along the positive or negative direction. Similarly x may be positive or negative.

Example 3.5

A particle starts with an initial velocity 2.5 m/s along the positive x direction and it accelerates uniformly at the rate 0.50 m/s^2 . (a) Find the distance travelled by it in the first two seconds. (b) How much time does it take to reach the velocity 7.5 m/s ? (c) How much distance will it cover in reaching the velocity 7.5 m/s ?

Solution : (a) We have,

$$\begin{aligned} x &= ut + \frac{1}{2}at^2 \\ &= (2.5 \text{ m/s})(2 \text{ s}) + \frac{1}{2}(0.50 \text{ m/s}^2)(2 \text{ s})^2 \\ &= 5.0 \text{ m} + 1.0 \text{ m} = 6.0 \text{ m}. \end{aligned}$$

Since the particle does not turn back it is also the distance travelled.

(b) We have,

$$v = u + at$$

$$\text{or, } 7.5 \text{ m/s} = 2.5 \text{ m/s} + (0.50 \text{ m/s}^2)t$$

$$\text{or, } t = \frac{7.5 \text{ m/s} - 2.5 \text{ m/s}}{0.50 \text{ m/s}^2} = 10 \text{ s}$$

(c) We have,

$$v^2 = u^2 + 2ax$$

$$\text{or, } (7.5 \text{ m/s})^2 = (2.5 \text{ m/s})^2 + 2(0.50 \text{ m/s}^2)x$$

$$\text{or, } x = \frac{(7.5 \text{ m/s})^2 - (2.5 \text{ m/s})^2}{2 \times 0.50 \text{ m/s}^2} = 50 \text{ m}.$$

Example 3.6

A particle having initial velocity u moves with a constant acceleration a for a time t . (a) Find the displacement of the particle in the last 1 second. (b) Evaluate it for $u = 5 \text{ m/s}$, $a = 2 \text{ m/s}^2$ and $t = 10 \text{ s}$.

Solution : (a) The position at time t is

$$s = ut + \frac{1}{2}at^2$$

The position at time $(t - 1 \text{ s})$ is

$$\begin{aligned} s' &= u(t - 1 \text{ s}) + \frac{1}{2}a(t - 1 \text{ s})^2 \\ &= ut - u(1 \text{ s}) + \frac{1}{2}at^2 - at(1 \text{ s}) + \frac{1}{2}a(1 \text{ s})^2 \end{aligned}$$

Thus, the displacement in the last 1 s is

$$\begin{aligned} s_t &= s - s' \\ &= u(1 \text{ s}) + at(1 \text{ s}) - \frac{1}{2}a(1 \text{ s})^2 \end{aligned}$$

$$\text{or, } s_t = u(1 \text{ s}) + \frac{a}{2}(2t - 1 \text{ s})(1 \text{ s}). \quad \dots (i)$$

(b) Putting the given values in (i)

$$\begin{aligned} s_t &= \left(5 \frac{\text{m}}{\text{s}}\right)(1 \text{ s}) + \frac{1}{2}\left(2 \frac{\text{m}}{\text{s}^2}\right)(2 \times 10 \text{ s} - 1 \text{ s})(1 \text{ s}) \\ &= 5 \text{ m} + \left(1 \frac{\text{m}}{\text{s}^2}\right)(19 \text{ s})(1 \text{ s}) \\ &= 5 \text{ m} + 19 \text{ m} = 24 \text{ m}. \end{aligned}$$

Sometimes, we are not careful in writing the units appearing with the numerical values of physical quantities. If we forget to write the unit of second in equation (i), we get,

$$s_t = u + \frac{a}{2}(2t - 1).$$

This equation is often used to calculate the displacement in the " t th second". However, as you can verify, different terms in this equation have different dimensions and hence the above equation is dimensionally incorrect. Equation (i) is the correct form which was used to solve part (b).

Also note that this equation gives the displacement of the particle in the last 1 second and not necessarily the distance covered in that second.

Freely Falling Bodies

A common example of motion in a straight line with constant acceleration is free fall of a body near the earth's surface. If air resistance is neglected and a body is dropped near the surface of the earth, it falls along a vertical straight line. The acceleration is in the vertically downward direction and its magnitude is almost constant if the height is small as compared with the radius of the earth (6400 km). This magnitude is approximately equal to 9.8 m/s^2 or 32 ft/s^2 and is denoted by the letter g .

If we take vertically upward as the positive Y -axis, acceleration is along the negative Y -axis and we write $a = -g$. The equation (3.12) to (3.14) may be written in this case as

$$\begin{aligned} v &= u - gt \\ y &= ut - \frac{1}{2}gt^2 \\ v^2 &= u^2 - 2gy. \end{aligned}$$

Here y is the y -coordinate (that is the height above the origin) at time t , u is the velocity in y direction at $t = 0$ and v is the velocity in y direction at time t . The position of the particle at $t = 0$ is $y = 0$.

Sometimes it is convenient to choose vertically downward as the positive Y -axis. Then $a = g$ and the equations (3.12) to (3.14) become

$$\begin{aligned} v &= u + gt \\ y &= ut + \frac{1}{2}gt^2 \\ v^2 &= u^2 + 2gy. \end{aligned}$$

Example 3.7

A ball is thrown up at a speed of 4.0 m/s . Find the maximum height reached by the ball. Take $g = 10 \text{ m/s}^2$.

Solution : Let us take vertically upward direction as the positive Y -axis. We have $u = 4.0 \text{ m/s}$ and $a = -10 \text{ m/s}^2$. At the highest point the velocity becomes zero. Using the formula.

$$v^2 = u^2 + 2ay,$$

$$0 = (4.0 \text{ m/s})^2 + 2(-10 \text{ m/s}^2)y$$

or, $y = \frac{16 \text{ m}^2/\text{s}^2}{20 \text{ m/s}^2} = 0.80 \text{ m}.$

3.7 MOTION IN A PLANE

If a particle is free to move in a plane, its position can be located with two coordinates. We choose the plane of motion as the X-Y plane. We choose a suitable instant as $t = 0$ and choose the origin at the place where the particle is situated at $t = 0$. Any two convenient mutually perpendicular directions in the X-Y plane are chosen as the X and Y-axes.

The position of the particle at a time t is completely specified by its coordinates (x, y) . The coordinates at time $t + \Delta t$ are $(x + \Delta x, y + \Delta y)$. Figure (3.8) shows the positions at t and $t + \Delta t$ as A and B respectively. The displacement during the time interval t to $t + \Delta t$ is

$$\vec{\Delta r} = \vec{AB} = \vec{AC} + \vec{CB}$$

$$= \Delta x \vec{i} + \Delta y \vec{j}$$

or, $\frac{\vec{\Delta r}}{\Delta t} = \frac{\Delta x}{\Delta t} \vec{i} + \frac{\Delta y}{\Delta t} \vec{j}.$

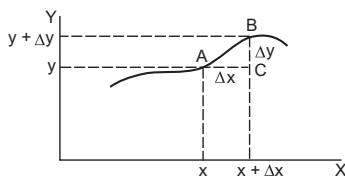


Figure 3.8

Taking limits as $\Delta t \rightarrow 0$

$$\vec{v} = \frac{dx}{dt} \vec{i} + \frac{dy}{dt} \vec{j}. \quad \dots (3.15)$$

Thus, we see that the x-component of the velocity is

$$v_x = \frac{dx}{dt} \quad \dots (3.16)$$

and the y-component is

$$v_y = \frac{dy}{dt}. \quad \dots (3.17)$$

Differentiating (3.15) with respect to time,

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt} \vec{i} + \frac{dv_y}{dt} \vec{j}$$

Thus, the acceleration has components

$$a_x = \frac{dv_x}{dt} \quad \dots (3.18)$$

$$\text{and} \quad a_y = \frac{dv_y}{dt}. \quad \dots (3.19)$$

We see that the x-coordinate, the x-component of velocity v_x and the x-component of acceleration a_x are related by

$$v_x = \frac{dx}{dt} \quad \text{and} \quad a_x = \frac{dv_x}{dt}.$$

These equations are identical to equations (3.9) and (3.10). Thus, if a_x is constant, integrating these equations we get

$$\left. \begin{aligned} v_x &= u_x + a_x t \\ x &= u_x t + \frac{1}{2} a_x t^2 \\ v_x^2 &= u_x^2 + 2a_x x \end{aligned} \right\} \quad \dots (3.20)$$

where u_x is the x-component of the velocity at $t = 0$. Similarly we have

$$v_y = \frac{dy}{dt} \quad \text{and} \quad a_y = \frac{dv_y}{dt}$$

and if a_y is constant,

$$\left. \begin{aligned} v_y &= u_y + a_y t \\ y &= u_y t + \frac{1}{2} a_y t^2 \\ v_y^2 &= u_y^2 + 2a_y y \end{aligned} \right\} \quad \dots (3.21)$$

The general scheme for the discussion of motion in a plane is therefore simple. The x-coordinate, the x-component of velocity and the x-component of acceleration are related by equations of straight line motion along the X-axis. Similarly the y-coordinate, the y-component of velocity and the y-component of acceleration are related by the equations of straight line motion along the Y-axis. The problem of motion in a plane is thus, broken up into two independent problems of straight line motion, one along the X-axis and the other along the Y-axis.

Example 3.8

A particle moves in the X-Y plane with a constant acceleration of 1.5 m/s^2 in the direction making an angle of 37° with the X-axis. At $t = 0$ the particle is at the origin and its velocity is 8.0 m/s along the X-axis. Find the velocity and the position of the particle at $t = 4.0 \text{ s}$.

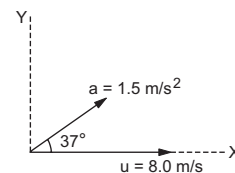


Figure 3.9

Solution : $a_x = (1.5 \text{ m/s}^2) (\cos 37^\circ)$
 $= (1.5 \text{ m/s}^2) \times \frac{4}{5} = 1.2 \text{ m/s}^2$
 and $a_y = (1.5 \text{ m/s}^2) (\sin 37^\circ)$
 $= (1.5 \text{ m/s}^2) \times \frac{3}{5} = 0.90 \text{ m/s}^2$.

The initial velocity has components

$$u_x = 8.0 \text{ m/s}$$

and $u_y = 0$

At $t = 0$, $x = 0$ and $y = 0$.

The x -component of the velocity at time $t = 4.0 \text{ s}$ is given by

$$\begin{aligned} v_x &= u_x + a_x t \\ &= 8.0 \text{ m/s} + (1.2 \text{ m/s}^2) (4.0 \text{ s}) \\ &= 8.0 \text{ m/s} + 4.8 \text{ m/s} = 12.8 \text{ m/s}. \end{aligned}$$

The y -component of velocity at $t = 4.0 \text{ s}$ is given by

$$\begin{aligned} v_y &= u_y + a_y t \\ &= 0 + (0.90 \text{ m/s}^2) (4.0 \text{ s}) = 3.6 \text{ m/s}. \end{aligned}$$

The velocity of the particle at $t = 4.0 \text{ s}$ is

$$\begin{aligned} v &= \sqrt{v_x^2 + v_y^2} = \sqrt{(12.8 \text{ m/s})^2 + (3.6 \text{ m/s})^2} \\ &= 13.3 \text{ m/s}. \end{aligned}$$

The velocity makes an angle θ with the X -axis where

$$\tan \theta = \frac{v_y}{v_x} = \frac{3.6 \text{ m/s}}{12.8 \text{ m/s}} = \frac{9}{32}.$$

The x -coordinate at $t = 4.0 \text{ s}$ is

$$\begin{aligned} x &= u_x t + \frac{1}{2} a_x t^2 \\ &= (8.0 \text{ m/s}) (4.0 \text{ s}) + \frac{1}{2} (1.2 \text{ m/s}^2) (4.0 \text{ s})^2 \\ &= 32 \text{ m} + 9.6 \text{ m} = 41.6 \text{ m}. \end{aligned}$$

The y -coordinate at $t = 4.0 \text{ s}$ is

$$\begin{aligned} y &= u_y t + \frac{1}{2} a_y t^2 \\ &= \frac{1}{2} (0.90 \text{ m/s}^2) (4.0 \text{ s})^2 \\ &= 7.2 \text{ m}. \end{aligned}$$

Thus, the particle is at $(41.6 \text{ m}, 7.2 \text{ m})$ at 4.0 s .

3.8 PROJECTILE MOTION

An important example of motion in a plane with constant acceleration is the projectile motion. When a particle is thrown obliquely near the earth's surface, it moves along a curved path. Such a particle is called a *projectile* and its motion is called *projectile motion*. We shall assume that the particle remains close to the surface of the earth and the air resistance is negligible. The acceleration of the particle is then almost

constant. It is in the vertically downward direction and its magnitude is g which is about 9.8 m/s^2 .

Let us first make ourselves familiar with certain terms used in discussing projectile motion. Figure (3.10) shows a particle projected from the point O with an initial velocity u at an angle θ with the horizontal. It goes through the highest point A and falls at B on the horizontal surface through O . The point O is called the *point of projection*, the angle θ is called the *angle of projection* and the distance OB is called the *horizontal range* or simply *range*. The total time taken by the particle in describing the path OAB is called the *time of flight*.

The motion of the projectile can be discussed separately for the horizontal and vertical parts. We take the origin at the point of projection. The instant

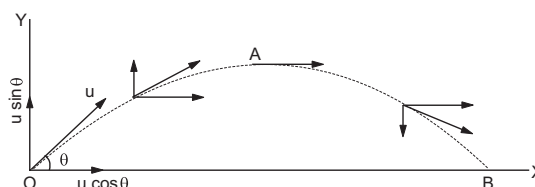


Figure 3.10

when the particle is projected is taken as $t = 0$. The plane of motion is taken as the X - Y plane. The horizontal line OX is taken as the X -axis and the vertical line OY as the Y -axis. Vertically upward direction is taken as the positive direction of the Y -axis.

We have $u_x = u \cos \theta$; $a_x = 0$
 $u_y = u \sin \theta$; $a_y = -g$.

Horizontal Motion

As $a_x = 0$, we have
 $v_x = u_x + a_x t = u_x = u \cos \theta$

and $x = u_x t + \frac{1}{2} a_x t^2 = u_x t = ut \cos \theta$.

As indicated in figure (3.10), the x -component of the velocity remains constant as the particle moves.

Vertical Motion

The acceleration of the particle is g in the downward direction. Thus, $a_y = -g$. The y -component of the initial velocity is u_y . Thus,

$$v_y = u_y - gt$$

and $y = u_y t - \frac{1}{2} gt^2$.

Also we have,

$$v_y^2 = u_y^2 - 2gy.$$

The vertical motion is identical to the motion of a particle projected vertically upward with speed $u \sin \theta$. The horizontal motion of the particle is identical to a particle moving horizontally with uniform velocity $u \cos \theta$.

Time of Flight

Consider the situation shown in figure (3.10). The particle is projected from the point O and reaches the same horizontal plane at the point B . The total time taken to reach B is the time of flight.

Suppose the particle is at B at a time t . The equation for horizontal motion gives

$$OB = x = ut \cos \theta$$

The y -coordinate at the point B is zero. Thus, from the equation of vertical motion,

$$y = ut \sin \theta - \frac{1}{2}gt^2$$

$$\text{or, } 0 = ut \sin \theta - \frac{1}{2}gt^2$$

$$\text{or, } t(u \sin \theta - \frac{1}{2}gt) = 0.$$

$$\text{Thus, either } t = 0 \text{ or, } t = \frac{2u \sin \theta}{g}.$$

Now $t = 0$ corresponds to the position O of the particle. The time at which it reaches B is thus,

$$T = \frac{2u \sin \theta}{g}. \quad \dots (3.22)$$

This is the time of flight.

Range

The distance OB is the horizontal range. It is the distance travelled by the particle in time $T = \frac{2u \sin \theta}{g}$.

By the equation of horizontal motion,

$$x = (u \cos \theta)t$$

$$\begin{aligned} \text{or, } OB &= (u \cos \theta) \left(\frac{2u \sin \theta}{g} \right) \\ &= \frac{2u^2 \sin \theta \cos \theta}{g} \\ &= \frac{u^2 \sin 2\theta}{g}. \quad \dots (3.23) \end{aligned}$$

Maximum Height Reached

At the maximum height (A in figure 3.10) the velocity of the particle is horizontal. The vertical component of velocity is thus, zero at the highest point. The maximum height is the y -coordinate of the particle when the vertical component of velocity becomes zero.

We have,

$$v_y = u_y - gt$$

$$= u \sin \theta - gt.$$

At the maximum height

$$0 = u \sin \theta - gt$$

$$\text{or, } t = \frac{u \sin \theta}{g}. \quad \dots (3.24)$$

The maximum height is

$$\begin{aligned} H &= u_y t - \frac{1}{2}gt^2 \\ &= (u \sin \theta) \left(\frac{u \sin \theta}{g} \right) - \frac{1}{2}g \left(\frac{u \sin \theta}{g} \right)^2 \\ &= \frac{u^2 \sin^2 \theta}{g} - \frac{1}{2} \frac{u^2 \sin^2 \theta}{g} \\ &= \frac{u^2 \sin^2 \theta}{2g}. \quad \dots (3.25) \end{aligned}$$

Equation (3.24) gives the time taken in reaching the maximum height. Comparison with equation (3.22) shows that it is exactly half the time of the flight. Thus, the time taken in ascending the maximum height equals the time taken in descending back to the same horizontal plane.

Example 3.9

A ball is thrown from a field with a speed of 12.0 m/s at an angle of 45° with the horizontal. At what distance will it hit the field again? Take $g = 10.0 \text{ m/s}^2$.

$$\begin{aligned} \text{Solution : The horizontal range} &= \frac{u^2 \sin 2\theta}{g} \\ &= \frac{(12 \text{ m/s})^2 \times \sin(2 \times 45^\circ)}{10 \text{ m/s}^2} \\ &= \frac{144 \text{ m}^2/\text{s}^2}{10.0 \text{ m/s}^2} = 14.4 \text{ m}. \end{aligned}$$

Thus, the ball hits the field at 14.4 m from the point of projection.

3.9 CHANGE OF FRAME

So far we have discussed the motion of a particle with respect to a given frame of reference. The frame can be chosen according to the convenience of the problem. The position $\vec{r}$, the velocity $\vec{v}$ and the acceleration $\vec{a}$ of a particle depend on the frame chosen. Let us see how can we relate the position, velocity and acceleration of a particle measured in two different frames.

Consider two frames of reference S and S' and suppose a particle P is observed from both the frames. The frames may be moving with respect to each other. Figure (3.11) shows the situation.

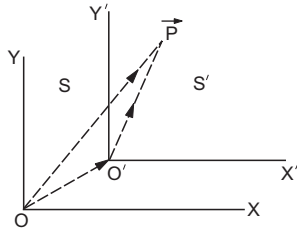


Figure 3.11

The position vector of the particle P with respect to the frame S is $\vec{r}_{P,S} = \vec{OP}$. The position vector of the particle with respect to the frame S' is $\vec{r}_{P,S'} = \vec{O'P}$. The position of the frame S' (the origin of frame S' in fact) with respect to the frame S is $\vec{OO'}$.

It is clear that

$$\vec{OP} = \vec{OO'} + \vec{O'P} = \vec{O'P} + \vec{OO'}$$

or, $\vec{r}_{P,S} = \vec{r}_{P,S'} + \vec{r}_{S',S}$... (3.26)

The position of the particle with respect to S is equal to the position of the particle with respect to S' plus the position of S' with respect to S .

If we differentiate equation (3.26) with respect to time, we get

$$\frac{d}{dt}(\vec{r}_{P,S}) = \frac{d}{dt}(\vec{r}_{P,S'}) + \frac{d}{dt}(\vec{r}_{S',S})$$

or, $\vec{v}_{P,S} = \vec{v}_{P,S'} + \vec{v}_{S',S}$... (3.27)

where $\vec{v}_{P,S}$ is the velocity of the particle with respect to S , $\vec{v}_{P,S'}$ is the velocity of the particle with respect to S' and $\vec{v}_{S',S}$ is the velocity of the frame S' with respect to S . The velocity of the particle with respect to S is equal to the velocity of the particle with respect to S' plus the velocity of S' with respect to S .

It is assumed that the meaning of time is same in both the frames. Similarly it is assumed that $\frac{d}{dt}$ has same meaning in both the frames. These assumptions are not correct if the velocity of one frame with respect to the other is so large that it is comparable to 3×10^8 m/s, or if one frame rotates with respect to the other. If the frames only translate with respect to each other with small velocity, the above assumptions are correct.

Equation (3.27) may be rewritten as

$$\vec{v}_{P,S'} = \vec{v}_{P,S} - \vec{v}_{S',S} \quad \dots (3.28)$$

Thus, if the velocities of two bodies (here the particle and the frame S') are known with respect to a common frame (here S) we can find the velocity of one body with respect to the other body. The velocity of body 1

with respect to the body 2 is obtained by subtracting the velocity of body 2 from the velocity of body 1.

When we say that the muzzle velocity of a bullet is 60 m/s we mean the velocity of the bullet with respect to the gun. If the gun is mounted in a train moving with a speed of 20 m/s with respect to the ground and the bullet is fired in the direction of the train's motion, its velocity with respect to the ground will be 80 m/s. Similarly, when we say that a swimmer can swim at a speed of 5 km/h we mean the velocity of the swimmer with respect to the water. If the water itself is flowing at 3 km/h with respect to the ground and the swimmer swims in the direction of the current, he or she will move at the speed of 8 km/h with respect to the ground.

Example 3.10

A swimmer can swim in still water at a rate 4.0 km/h. If he swims in a river flowing at 3.0 km/h and keeps his direction (with respect to water) perpendicular to the current, find his velocity with respect to the ground.

Solution : The velocity of the swimmer with respect to water is $\vec{v}_{S,R} = 4.0$ km/h in the direction perpendicular to the river. The velocity of river with respect to the ground is $\vec{v}_{R,G} = 3.0$ km/h along the length of the river. The velocity of the swimmer with respect to the ground is $\vec{v}_{S,G}$ where

$$\vec{v}_{S,G} = \vec{v}_{S,R} + \vec{v}_{R,G}$$

Figure (3.12) shows the velocities. It is clear that,

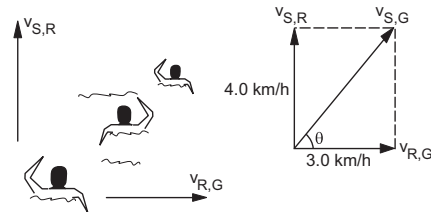


Figure 3.12

$$v_{S,G} = \sqrt{(4.0 \text{ km/h})^2 + (3.0 \text{ km/h})^2}$$

$$= 5.0 \text{ km/h}$$

The angle θ made with the direction of flow is

$$\tan \theta = \frac{4.0 \text{ km/h}}{3.0 \text{ km/h}} = \frac{4}{3}$$

Example 3.11

A man is walking on a level road at a speed of 3.0 km/h. Rain drops fall vertically with a speed of 4.0 km/h. Find the velocity of the raindrops with respect to the man.

Solution : We have to find the velocity of raindrops with respect to the man. The velocity of the rain as well as the velocity of the man are given with respect to the street. We have

$$\vec{v}_{\text{rain, man}} = \vec{v}_{\text{rain, street}} - \vec{v}_{\text{man, street}}.$$

Figure (3.13) shows the velocities.

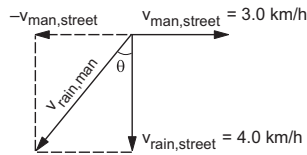


Figure 3.13

It is clear from the figure that

$$v_{\text{rain, man}} = \sqrt{(4.0 \text{ km/h})^2 + (3.0 \text{ km/h})^2} = 5.0 \text{ km/h}.$$

The angle with the vertical is θ , where

$$\tan \theta = \frac{3.0 \text{ km/h}}{4.0 \text{ km/h}} = \frac{3}{4}.$$

Thus, the rain appears to fall at an angle $\tan^{-1}(3/4)$ with the speed 5.0 km/h as viewed by the man.

The relation between the accelerations measured from two frames can be obtained by differentiating equation (3.27) with respect to time.

We have,

$$\frac{d}{dt}(\vec{v}_{P,S}) = \frac{d}{dt}(\vec{v}_{P,S'}) + \frac{d}{dt}(\vec{v}_{S',S})$$

or, $\vec{a}_{P,S} = \vec{a}_{P,S'} + \vec{a}_{S',S} \quad \dots (3.29)$

If S' moves with respect to S at a uniform velocity, $\vec{a}_{S',S} = 0$ and so

$$\vec{a}_{P,S} = \vec{a}_{P,S'}.$$

If two frames are moving with respect to each other with uniform velocity, acceleration of a body is same in both the frames.

Worked Out Examples

1. A man walks at a speed of 6 km/hr for 1 km and 8 km/hr for the next 1 km. What is his average speed for the walk of 2 km ?

Solution : Distance travelled is 2 km.

$$\begin{aligned} \text{Time taken} &= \frac{1 \text{ km}}{6 \text{ km/hr}} + \frac{1 \text{ km}}{8 \text{ km/hr}} \\ &= \left(\frac{1}{6} + \frac{1}{8}\right) \text{ hr} = \frac{7}{24} \text{ hr}. \end{aligned}$$

$$\begin{aligned} \text{Average speed} &= \frac{2 \text{ km} \times 24}{7 \text{ hr}} = \frac{48}{7} \text{ km/hr} \\ &\approx 7 \text{ km/hr}. \end{aligned}$$

2. The I.Sc. lecture theatre of a college is 40 ft wide and has a door at a corner. A teacher enters at 12:00 noon through the door and makes 10 rounds along the 40 ft wall back and forth during the period and finally leaves the class-room at 12:50 p.m. through the same door. Compute his average speed and average velocity.

Solution : Total distance travelled in 50 minutes = 800 ft.

$$\text{Average speed} = \frac{800}{50} \text{ ft/min} = 16 \text{ ft/min}.$$

At 12:00 noon he is at the door and at 12:50 pm he is again at the same door.

The displacement during the 50 min interval is zero.

Average velocity = zero.

3. The position of a particle moving on X-axis is given by

$$x = At^3 + Bt^2 + Ct + D.$$

The numerical values of A, B, C, D are 1, 4, -2 and 5

respectively and SI units are used. Find (a) the dimensions of A, B, C and D , (b) the velocity of the particle at $t = 4$ s, (c) the acceleration of the particle at $t = 4$ s, (d) the average velocity during the interval $t = 0$ to $t = 4$ s, (e) the average acceleration during the interval $t = 0$ to $t = 4$ s.

Solution : (a) Dimensions of x, At^3, Bt^2, Ct and D must be identical and in this case each is length. Thus,

$$[At^3] = L, \text{ or, } [A] = LT^{-3}$$

$$[Bt^2] = L, \text{ or, } [B] = LT^{-2}$$

$$[Ct] = L, \text{ or, } [C] = LT^{-1}$$

$$\text{and } [D] = L.$$

$$(b) \quad x = At^3 + Bt^2 + Ct + D$$

$$\text{or, } v = \frac{dx}{dt} = 3At^2 + 2Bt + C.$$

Thus, at $t = 4$ s, the velocity

$$\begin{aligned} &= 3(1 \text{ m/s}^3)(16 \text{ s}^2) + 2(4 \text{ m/s}^2)(4 \text{ s}) + (-2 \text{ m/s}) \\ &= (48 + 32 - 2) \text{ m/s} = 78 \text{ m/s}. \end{aligned}$$

$$(c) \quad v = 3At^2 + 2Bt + C$$

$$\text{or, } a = \frac{dv}{dt} = 6At + 2B.$$

$$\text{At } t = 4 \text{ s, } a = 6(1 \text{ m/s}^3)(4 \text{ s}) + 2(4 \text{ m/s}^2) = 32 \text{ m/s}^2.$$

$$(d) \quad x = At^3 + Bt^2 + Ct + D.$$

Position at $t = 0$ is $x = D = 5$ m.

Position at $t = 4$ s is

$$\begin{aligned} &(1 \text{ m/s}^3)(64 \text{ s}^3) + (4 \text{ m/s}^2)(16 \text{ s}^2) - (2 \text{ m/s})(4 \text{ s}) + 5 \text{ m} \\ &= (64 + 64 - 8 + 5) \text{ m} = 125 \text{ m}. \end{aligned}$$

CHAPTER 4

THE FORCES

4.1 INTRODUCTION

Force is a word which we have all heard about. When you push or pull some object you exert a force on it. If you push a body you exert a force away from yourself; when you pull, you exert a force toward yourself. When you hold a heavy block in your hand you exert a large force; when you hold a light block, you exert a small force.

Can nonliving bodies exert a force? Yes, they can. If we stand in a great storm, we feel that the wind is exerting a force on us. When we suspend a heavy block from a rope, the rope holds the block just as a man can hold it in air. When we comb our dry hair and bring the comb close to small pieces of paper, the pieces jump to the comb. The comb has attracted the paper pieces i.e. the comb has exerted force on the pieces. When a cork is dipped in water it comes to the surface; if we want to keep it inside water, we have to push it downward. We say that water exerts a force on the cork in the upward direction.

The SI unit for measuring the force is called a *newton*. Approximately, it is the force needed to hold a body of mass 102 g near the earth's surface. An accurate quantitative definition can be framed using Newton's laws of motion to be studied in the next chapter.

Force is an interaction between two objects. Force is exerted by an object *A* on another object *B*. For any force you may ask two questions, (i) who exerted this force and (ii) on which object was this force exerted? Thus, when a block is kept on a table, the table exerts a force on the block to hold it.

Force is a vector quantity and if more than one forces act on a particle we can find the resultant force using the laws of vector addition. Note that in all the examples quoted above, if a body *A* exerts a force on *B*, the body *B* also exerts a force on *A*. Thus, the table exerts a force on the block to hold it and the block exerts a force on the table to press it down. When a heavy block is suspended by a rope, the rope exerts a

force on the block to hold it and the block exerts a force on the rope to make it tight and stretched. In fact these are a few examples of Newton's third law of motion which may be stated as follows.

Newton's Third Law of Motion

If a body A exerts a force $\vec{F}$ on another body B, then B exerts a force $-\vec{F}$ on A, the two forces acting along the line joining the bodies.

The two forces $\vec{F}$ and $-\vec{F}$ connected by Newton's third law are called *action-reaction pair*. Any one may be called 'action' and the other 'reaction'.

We shall discuss this law in greater detail in the next chapter.

The various types of forces in nature can be grouped in four categories:

- (a) Gravitational, (b) Electromagnetic,
- (c) Nuclear and (d) Weak.

4.2 GRAVITATIONAL FORCE

Any two bodies attract each other by virtue of their masses. The force of attraction between two point masses is $F = G \frac{m_1 m_2}{r^2}$, where m_1 and m_2 are the masses

of the particles and r is the distance between them. G is a universal constant having the value $6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$. To find the gravitational force on an extended body by another such body, we have to write the force on each particle of the 1st body by all the particles of the second body and then we have to sum up vectorially all the forces acting on the first body. For example, suppose each body contains just three particles, and let $\vec{F}_{ij}$ denote the force on the i th particle of the first body due to the j th particle of the second body. To find the resultant force on the first body (figure 4.1), we have to add the following 9 forces:

$$\vec{F}_{11}, \vec{F}_{12}, \vec{F}_{13}, \vec{F}_{21}, \vec{F}_{22}, \vec{F}_{23}, \vec{F}_{31}, \vec{F}_{32}, \vec{F}_{33}.$$

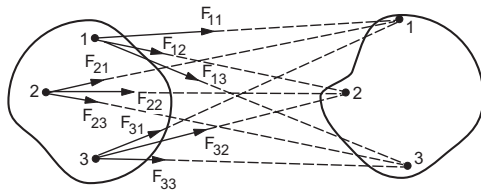


Figure 4.1

For large bodies having a large number of particles, we have to add quite a large number of forces. If the bodies are assumed continuous (a good approximation in our course), one has to go through the integration process for the infinite summation involved. However, the integration yields a particularly simple result for a special case which is of great practical importance and we quote it below. The proof of this result will be given in a later chapter.

The gravitational force exerted by a spherically symmetric body of mass m_1 on another such body of mass m_2 kept outside the first body is $G \frac{m_1 m_2}{r^2}$, where r is the distance between the centres of the two bodies. Thus, for the calculation of gravitational force between two spherically symmetric bodies, they can be treated as point masses placed at their centres.

Gravitational Force on Small Bodies by the Earth

The force of attraction exerted by the earth on other objects is called *gravity*. Consider the earth to be a homogeneous sphere of radius R and mass M . The values of R and M are roughly 6400 km and 6×10^{24} kg respectively. Assuming that the earth is spherically symmetric, the force it exerts on a particle of mass m kept near its surface is by the previous result, $F = G \frac{Mm}{R^2}$. The direction of this force is towards the centre of the earth which is called the *vertically downward* direction.

The same formula is valid to a good approximation even if we have a body of some other shape instead of a particle, provided the body is very small in size as compared to the earth. The quantity $G \frac{M}{R^2}$ is a constant and has the dimensions of acceleration. It is called the *acceleration due to gravity*, and is denoted by the letter g (a quantity much different from G). Its value is approximately 9.8 m/s^2 . For simplicity of calculations we shall often use $g = 10 \text{ m/s}^2$. We shall find in the next chapter that all bodies falling towards earth (remaining all the time close to the earth's surface) have this particular value of acceleration and hence the name acceleration due to gravity. Thus, the force exerted by the earth on a small body of mass m , kept

near the earth's surface is mg in the vertically downward direction.

The gravitational constant G is so small that the gravitational force becomes appreciable only if at least one of the two bodies has a large mass. To have an idea of the magnitude of gravitational forces in practical life, consider two small bodies of mass 10 kg each, separated by 0.5 m. The gravitational force is

$$F = \frac{6.7 \times 10^{-11} \text{ N-m}^2/\text{kg}^2 \times 10^2 \text{ kg}^2}{0.25 \text{ m}^2} \\ = 2.7 \times 10^{-8} \text{ N}$$

a force needed to hold about 3 microgram. In many of the situations we encounter, it is a good approximation to neglect all the gravitational forces other than that exerted by the earth.

4.3 ELECTROMAGNETIC (EM) FORCE

Over and above the gravitational force $G \frac{m_1 m_2}{r^2}$, the particles may exert upon each other electromagnetic forces. If two particles having charges q_1 and q_2 are at rest with respect to the observer, the force between them has a magnitude

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

where $\epsilon = 8.85419 \times 10^{-12} \text{ C}^2/\text{N-m}^2$ is a constant. The quantity $\frac{1}{4\pi\epsilon_0}$ is $9.0 \times 10^9 \frac{\text{N-m}^2}{\text{C}^2}$.

This is called *Coulomb force* and it acts along the line joining the particles. If q_1 and q_2 are of same nature (both positive or both negative), the force is repulsive otherwise it is attractive. It is this force which is responsible for the attraction of small paper pieces when brought near a recently used comb. The electromagnetic force between moving charged particles is comparatively more complicated and contains terms other than the Coulomb force.

Ordinary matter is composed of electrons, protons and neutrons. Each electron has 1.6×10^{-19} coulomb of negative charge and each proton has an equal amount of positive charge. In atoms, the electrons are bound by the electromagnetic force acting on them due to the protons. The atoms combine to form molecules due to the electromagnetic forces. A lot of atomic and molecular phenomena result from electromagnetic forces between the subatomic particles (electrons, protons, charged mesons, etc.).

Apart from the atomic and molecular phenomena, the electromagnetic forces show up in many forms in

daily experience. Some examples having practical importance given below.

(a) Forces between Two Surfaces in Contact

When we put two bodies in contact with each other, the atoms at the two surfaces come close to each other. The charged constituents of the atoms of the two bodies exert great forces on each other and a measurable force results out of it. We say that the two bodies in contact exert forces on each other. When you place a book on a table, the table exerts an upward force on the book to hold it. This force comes from the electromagnetic forces acting between the atoms and molecules of the surface of the book and of the table.

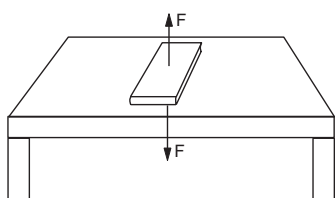


Figure 4.2

Generally, the forces between the two objects in contact are along the common normal (perpendicular) to the surfaces of contact and is that of a push or repulsion. Thus, the table pushes the book away from it (i.e., upward) and the book pushes the table downward (again away from it).

However, the forces between the two bodies in contact may have a component parallel to the surface of contact. This component is known as *friction*. We assume existence of frictionless surfaces which can exert forces only along the direction perpendicular to them. The bodies with smooth surfaces can exert only small amount of forces parallel to the surface and hence are close to frictionless surface. Thus, it is difficult to stay on a smooth metallic lamp-post, because it cannot exert enough vertical force and so it will not hold you there. The same is not true if you try to stay on the trunk of a tree which is quite rough. We shall often use the word smooth to mean frictionless.

The contact forces obey Newton's third law. Thus the book in figure (4.2) exerts a downward force F on the table to press it down and the table exerts an equal upward force F on the book to hold it there. When you stay on the trunk of a tree, it exerts a frictional upward force (frictional force because it is parallel to the surface of the tree) on you to hold you there, and you exert an equal frictional downward force on the tree.

(b) Tension in a String or a Rope

In a tug of war, two persons hold the two ends of a rope and try to pull the rope on their respective sides. The rope becomes tight and its length is slightly increased. In many situations this increase is very small and goes undetected. Such a stretched rope is said to be in a state of tension.

Similarly, if a heavy block hangs from a ceiling by a string, the string is in a state of tension. The electrons and protons of the string near the lower end exert forces on the electrons and protons of the block and the resultant of these forces is the force exerted by the string on the block. It is the resultant of these electromagnetic forces which supports the block and prevents it from falling. A string or rope under tension exerts electromagnetic forces on the bodies attached at the two ends to *pull* them.

(c) Force due to a Spring

When a metallic wire is coiled it becomes a spring. The straight line distance between the ends of a spring is called its length. If a spring is placed on a horizontal surface with no horizontal force on it, its length is called the *natural length*. Every spring has its own natural length. The spring can be stretched to increase its length and it can be compressed to decrease its length. When a spring is stretched, it pulls the bodies attached to its ends and when compressed, it pushes the bodies attached to its ends. If the extension or the compression is not too large, the force exerted by the spring is proportional to the change in its length. Thus, if the spring has a length x and its natural length is x_0 the magnitude of the force exerted by it will be

$$F = k|x - x_0| = k|\Delta x|.$$

If the spring is extended, the force will be directed towards its centre and if compressed, it will be directed away from the centre. The proportionality constant k , which is the force per unit extension or compression, is called the *spring constant* of the spring. This force again comes into picture due to the electromagnetic forces between the atoms of the material.

The macroscopic bodies which we have to generally deal with are electrically neutral. Hence two bodies not in contact do not exert appreciable electromagnetic forces. The forces between the charged particles of the first body and those of the second body have both attractive and repulsive nature and hence they largely cancel each other. This is not the case with gravitational forces. The gravitational forces between the particles of one body and those of the other body are all attractive and hence they add to give an appreciable gravitational force in many cases. Thus, the gravitational force between the earth and a 1 kg

block kept 100 m above the earth's surface is about 9.8 N whereas the electromagnetic force between the earth and this block is almost zero even though both these bodies contain a very large number of charged particles, the electrons and the protons.

Example 4.1

Suppose the exact charge neutrality does not hold in a world and the electron has a charge 1% less in magnitude than the proton. Calculate the Coulomb force acting between two blocks of iron each of mass 1 kg separated by a distance of 1 m. Number of protons in an iron atom = 26 and 58 kg of iron contains 6×10^{26} atoms.

Solution : Each atom of iron will have a net positive charge $26 \times 0.01 \times 1.6 \times 10^{-19}$ C on it in the assumed world. The total positive charge on a 1 kg block would be

$$\frac{6 \times 10^{26}}{58} \times 26 \times 1.6 \times 10^{-21} \text{ C}$$

$$= 4.3 \times 10^5 \text{ C.}$$

The Coulomb force between the two blocks is

$$= \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} = \frac{9.0 \times 10^9 \text{ N-m}^2/\text{C}^2 \times (4.3 \times 10^5 \text{ C})^2}{(1 \text{ m})^2}$$

$$= 9 \times 10^9 \times 18.49 \times 10^{10} \text{ N}$$

$$= 1.7 \times 10^{21} \text{ N.}$$

A tremendous force indeed !

4.4 NUCLEAR FORCES

Each atom contains a certain number of protons and neutrons in its nucleus. The nucleus occupies a volume of about 10^{-44} m^3 whereas the atom itself has a volume of about 10^{-23} m^3 . Thus, the nucleus occupies only $1/10^{21}$ of the volume of the atom. Yet it contains about 99.98% of the mass of the atom. The atomic nucleus of a non-radioactive element is a stable particle. For example, if both the electrons are removed from a helium atom, we get the bare nucleus of helium which is called an *alpha particle*. The alpha particle is a stable object and once created it can remain intact until it is not made to interact with other objects.

An alpha particle contains two protons and two neutrons. The protons will repel each other due to the Coulomb force and will try to break the nucleus. Neutrons will be silent spectators in this electromagnetic drama (Remember, neutron is an uncharged particle). Then, why does the Coulomb force fail to break the nucleus? Can it be the gravitational attractive force which keeps the nucleus bound? All the protons and the neutrons will take part in this attraction, but if calculated, the gravitational

attraction will turn out to be totally negligible as compared to the Coulomb repulsion.

In fact, a third kind of force, altogether different and over and above the gravitational and electromagnetic force, is operating here. These forces are called *Nuclear forces* and are exerted only if the interacting particles are protons or neutrons or both. (There are some more cases where this force operates but we shall not deal with them.) These forces are largely attractive, but are short ranged. The forces are much weaker than the Coulomb force if the separation between the particles is more than say 10^{-14} m. But for smaller separation ($\approx 10^{-15}$ m) the nuclear force is much stronger than the Coulomb force and being attractive it holds the nucleus stable.

Being short ranged, these forces come into picture only if the changes within the nucleus are discussed. As bare nuclei are less frequently encountered in daily life, one is generally unaware of these forces. Radioactivity, nuclear energy (fission, fusion) etc. result from nuclear forces.

4.5 WEAK FORCES

Yet another kind of forces is encountered when reactions involving protons, electrons and neutrons take place. A neutron can change itself into a proton and simultaneously emit an electron and a particle called *antineutrino*. This is called β^- decay. Never think that a neutron is made up of a proton, an electron and an antineutrino. A proton can also change into neutron and simultaneously emit a positron (and a neutrino). This is called β^+ decay. The forces responsible for these changes are different from gravitational, electromagnetic or nuclear forces. Such forces are called weak forces. The range of weak forces is very small, in fact much smaller than the size of a proton or a neutron. Thus, its effect is experienced inside such particles only.

4.6 SCOPE OF CLASSICAL PHYSICS

The behaviour of all the bodies of linear sizes greater than 10^{-6} m are adequately described on the basis of relatively a small number of very simple laws of nature. These laws are the Newton's laws of motion, Newton's law of gravitation, Maxwell's electromagnetism, Laws of thermodynamics and the Lorentz force. The principles of physics based on them is called the *classical physics*. The formulation of classical physics is quite accurate for heavenly bodies like the sun, the earth, the moon etc. and is equally good for the behaviour of grains of sand and the raindrops. However, for the subatomic particles much smaller

than 10^{-6} m (such as atoms, nuclei etc.) these rules do not work well. The behaviour of such particles is governed by *quantum physics*. In fact, at such short dimensions the very concept of “particle” breaks down. The perception of the nature is altogether different at this scale. The validity of classical physics also depends on the velocities involved. The classical mechanics as formulated by Newton has to be considerably changed when velocities comparable to 3×10^8 m/s are involved. This is the speed of light in vacuum and is the upper limit of speed which material particle can ever reach. No matter how great and how long you apply a force, you can never get a particle going with a speed greater than 3×10^8 m/s. The mechanics of particles moving with these large velocities is known as *relativistic mechanics* and was formulated by Einstein in 1905.

Thus, classical physics is a good description of the nature if we are concerned with the particles of linear

size $> 10^{-6}$ m moving with velocities $< 10^8$ m/s. In a major part of this book, we shall work within these restrictions and hence learn the techniques of classical physics. The size restriction automatically excludes any appreciable effects of nuclear or weak forces and we need to consider only the gravitational and electromagnetic forces. We might consider the subatomic particles here and there but shall assume the existence of gravitational and electromagnetic forces only and that classical physics is valid for these particles. The results arrived at by our analysis may only be approximately true because we shall be applying the laws which are not correct in that domain. But even that may play an important role in the understanding of nature. We shall also assume that the Newton's third law is valid for the forces which we shall be dealing with. In the final chapters we shall briefly discuss quantum physics and some of its important consequences.

Worked Out Examples

1. Figure (4-W1) shows two hydrogen atoms. Show on a separate diagram all the electric forces acting on different particles of the system.

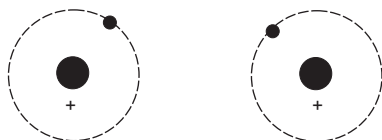


Figure 4-W1

Solution : Each particle exerts electric forces on the remaining three particles. Thus there exist $4 \times 3 = 12$ forces in all. Figure (4-W2) shows them.

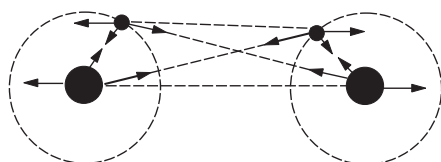


Figure 4-W2

2. Figure (4-W3) shows two rods each of length l placed side by side, with their facing ends separated by a distance a . Charges $+q$, $-q$ reside on the rods as shown. Calculate the electric force on the rod A due to the rod B. Discuss the cases when $l \gg a$, $a \gg l$.

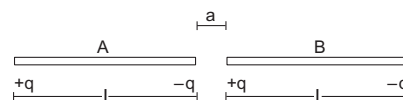


Figure 4-W3

Solution : The force on the rod A due to the charge $+q$ of the rod B

$$= -\frac{q^2}{4\pi\epsilon_0(l+a)^2} + \frac{q^2}{4\pi\epsilon_0 a^2}$$

towards right. The force on this rod due to the charge $-q$

$$= \frac{q^2}{4\pi\epsilon_0(2l+a)^2} - \frac{q^2}{4\pi\epsilon_0(l+a)^2}$$

towards right.

The resultant force on the rod is

$$F = \frac{q^2}{4\pi\epsilon_0} \left[\frac{1}{a^2} - \frac{2}{(l+a)^2} + \frac{1}{(2l+a)^2} \right] \text{ towards right.}$$

If $l \gg a$, the last two terms in the square bracket are negligible as compared to the first term. Then,

$$F \approx \frac{q^2}{4\pi\epsilon_0 a^2}.$$

If $a \gg l$

$$F \approx \frac{q^2}{4\pi\epsilon_0} \left[\frac{1}{a^2} - \frac{2}{a^2} + \frac{1}{a^2} \right] \approx 0.$$

CHAPTER 5

NEWTON'S LAWS OF MOTION

Newton's laws of motion are of central importance in classical physics. A large number of principles and results may be derived from Newton's laws. The first two laws relate to the type of motion of a system that results from a given set of forces. These laws may be interpreted in a variety of ways and it is slightly uninteresting and annoying at the outset to go into the technical details of the interpretation. The precise definitions of mass, force and acceleration should be given before we relate them. And these definitions themselves need use of Newton's laws. Thus, these laws turn out to be definitions to some extent. We shall assume that we know how to assign mass to a body, how to assign the magnitude and direction to a force and how to measure the acceleration with respect to a given frame of reference. Some discussions of these aspects were given in the previous chapters. The development here does not follow the historical track these laws have gone through, but are explained to make them simple to apply.

5.1 FIRST LAW OF MOTION

If the (vector) sum of all the forces acting on a particle is zero then and only then the particle remains unaccelerated (i.e., remains at rest or moves with constant velocity).

If the sum of all the forces on a given particle is $\vec{F}$ and its acceleration is $\vec{a}$, the above statement may also be written as

$\vec{a} = 0$ if and only if $\vec{F} = 0$.

Thus, if the sum of the forces acting on a particle is known to be zero, we can be sure that the particle is unaccelerated, or if we know that a particle is unaccelerated, we can be sure that the sum of the forces acting on the particle is zero.

However, the concept of rest, motion or acceleration is meaningful only when a frame of reference is specified. Also the acceleration of the

particle is, in general, different when measured from different frames. Is it possible then, that the first law is valid in all frames of reference?

Let us consider the situation shown in figure (5.1). An elevator cabin falls down after the cable breaks. The cabin and all the bodies fixed in the cabin are accelerated with respect to the earth and the acceleration is about 9.8 m/s^2 in the downward direction.

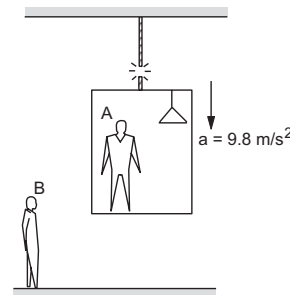


Figure 5.1

Consider the lamp in the cabin. The forces acting on the lamp are (a) the gravitational force W by the earth and (b) the electromagnetic force T (tension) by the rope. The direction of W is downward and the direction of T is upward. The sum is $(W - T)$ downward.

Measure the acceleration of the lamp from the frame of reference of the cabin. The lamp is at rest. The acceleration of the lamp is zero. The person A who measured this acceleration is a learned one and uses Newton's first law to conclude that the sum of the forces acting on the particle is zero, i.e.,

$$W - T = 0 \text{ or } W = T.$$

Instead, if we measure the acceleration from the ground, the lamp has an acceleration of 9.8 m/s^2 . Thus, $a \neq 0$ and hence the person B who measured this acceleration, concludes from Newton's first law that the sum of the forces is not zero. Thus, $W - T \neq 0$ or $W \neq T$. If A measures acceleration and applies the first

law he gets $W = T$. If B measures acceleration and applies the same first law, he gets $W \neq T$. Both of them cannot be correct simultaneously as W and T can be either equal or unequal. At least one of the two frames is a bad frame and one should not apply the first law in that frame.

There are some frames of reference in which Newton's first law is valid. Measure acceleration from such a frame and you are allowed to say that " $\vec{a} = 0$ if and only if $\vec{F} = 0$ ". But there are frames in which Newton's first law is not valid. You may find that even if the sum of the forces is not zero, the acceleration is still zero. Or you may find that the sum of the forces is zero, yet the particle is accelerated. So the validity of Newton's first law depends on the frame of reference from which the observer measures the state of rest, motion and acceleration of the particle.

A frame of reference in which Newton's first law is valid is called an *inertial frame of reference*. A frame in which Newton's first law is not valid is called a *noninertial frame of reference*.

Newton's first law, thus, reduces to a definition of inertial frame. Why do we call it a law then? Suppose after going through this lesson, you keep the book on your table fixed rigidly with the earth (figure 5.2).

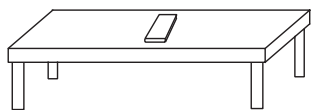


Figure 5.2

The book is at rest with respect to the earth. The acceleration of the book with respect to the earth is zero. The forces on the book are (a) the gravitational force $\vec{W}$ exerted by the earth and (b) the contact force $\vec{N}$ by the table. Is the sum of $\vec{W}$ and $\vec{N}$ zero? A very accurate measurement will give the answer "No". The sum of the forces is not zero although the book is at rest. The earth is not strictly an inertial frame. However, the sum is not too different from zero and we can say that the earth is an inertial frame of reference to a good approximation. Thus, for routine affairs, " $\vec{a} = 0$ if and only if $\vec{F} = 0$ " is true in the earth frame of reference. This fact was identified and formulated by Newton and is known as Newton's *first law*. If we restrict that all measurements will be made from the earth frame, indeed it becomes a law. If we try to universalise this to different frames, it becomes a definition. We shall assume that unless stated otherwise, we are working from an inertial frame of reference.

Example 5.1

A heavy particle of mass 0.50 kg is hanging from a string fixed with the roof. Find the force exerted by the string on the particle (Figure 5.3). Take $g = 9.8 \text{ m/s}^2$.

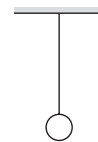


Figure 5.3

Solution : The forces acting on the particle are

- (a) pull of the earth, $0.50 \text{ kg} \times 9.8 \text{ m/s}^2 = 4.9 \text{ N}$, vertically downward
- (b) pull of the string, T vertically upward.

The particle is at rest with respect to the earth (which we assume to be an inertial frame). Hence, the sum of the forces should be zero. Therefore, T is 4.9 N acting vertically upward.

Inertial Frames other than Earth

Suppose S is an inertial frame and S' a frame moving uniformly with respect to S . Consider a particle P having acceleration $\vec{a}_{P,S}$ with respect to S and $\vec{a}_{P,S'}$ with respect to S' .

We know that,

$$\vec{a}_{P,S} = \vec{a}_{P,S'} + \vec{a}_{S',S}.$$

As S' moves uniformly with respect to S ,

$$\vec{a}_{S',S} = 0.$$

Thus,

$$\vec{a}_{P,S} = \vec{a}_{P,S'} \quad \dots (i)$$

Now S is an inertial frame. So from definition, $\vec{a}_{P,S} = 0$, if and only if $\vec{F} = 0$ and hence, from (i), $\vec{a}_{P,S'} = 0$ if and only if $\vec{F} = 0$.

Thus, S' is also an inertial frame. We arrive at an important result : *All frames moving uniformly with respect to an inertial frame are themselves inertial.* Thus, a train moving with uniform velocity with respect to the ground, a plane flying with uniform velocity with respect to a highway, etc., are examples of inertial frames. The sum of the forces acting on a suitcase kept on the shelf of a ship sailing smoothly and uniformly on a calm sea is zero.

5.2 SECOND LAW OF MOTION

The acceleration of a particle as measured from an inertial frame is given by the (vector) sum of all the forces acting on the particle divided by its mass.

In symbols : $\vec{a} = \vec{F}/m$ or, $\vec{F} = m\vec{a}$ (5.2)

The inertial frame is already defined by the first law of motion. A force $\vec{F}$ acting on a particle of mass m produces an acceleration $\vec{F}/m$ in it with respect to an inertial frame. This is a law of nature. If the force ceases to act at some instant, the acceleration becomes zero at the same instant. In equation (5.2) $\vec{a}$ and $\vec{F}$ are measured at the same instant of time.

5.3 WORKING WITH NEWTON'S FIRST AND SECOND LAW

Newton's laws refer to a particle and relate the forces acting on the particle with its acceleration and its mass. Before attempting to write an equation from Newton's law, we should very clearly understand which particle we are considering. In any practical situation, we deal with extended bodies which are collection of a large number of particles. The laws as stated above may be used even if the object under consideration is an extended body, provided each part of this body has the same acceleration (in magnitude and direction). A systematic algorithm for writing equations from Newton's laws is as follows :

Step 1 : Decide the System

The first step is to decide the system on which the laws of motion are to be applied. The system may be a single particle, a block, a combination of two blocks one kept over the other, two blocks connected by a string, a piece of string etc. The only restriction is that all parts of the system should have identical acceleration.

Consider the situation shown in figure (5.4). The block B does not slip over A , the disc D slides over the string and all parts of the string are tight.

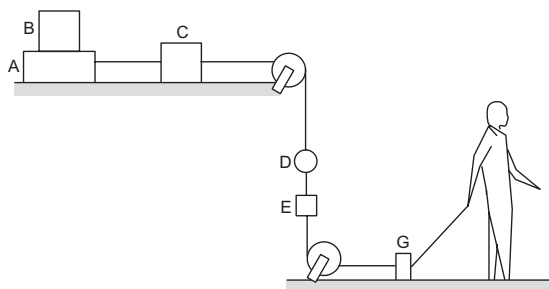


Figure 5.4

A and B move together. C is not in contact with A or B . But as the length of the string between A and C does not change, the distance moved by C in any

time interval is same as that by A . The same is true for G . The distance moved by G in any time interval is same as that by A , B or C . The direction of motion is also the same for A , B , C and G . They have identical accelerations. We can take any of these blocks as a system or any combination of the blocks from these as a system. Some of the examples are (A) , (B) , $(A + B)$, $(B + C)$, $(A + B + C)$, $(C + G)$, $(A + C + G)$, $(A + B + C + G)$ etc. The distance covered by E is also the same as the distance covered by G but their directions are different. E moves in a vertical line whereas G in a horizontal line. $(E + G)$ should not be taken as a system. At least at this stage we are unable to apply Newton's law treating $E + G$ as a single particle. As the disc D slides over the string the distance covered by D is not equal to that by E in the same time interval. We should not treat $D + E$ as a system. *Think carefully.*

Step 2 : Identify the Forces

Once the system is decided, make a list of the forces acting *on* the system due to all the objects other than the system. Any force applied *by* the system should not be included in the list of the forces.

Consider the situation shown in figure (5.5). The boy stands on the floor balancing a heavy load on his head. The load presses the boy, the boy pushes the load upward the boy presses the floor downward, the floor pushes the boy upward, the earth attracts the load downward, the load attracts the earth upward, the boy attracts the earth upward and the earth attracts the boy downward. There are many forces operating in this world. Which of these forces should we include in the list of forces ?



Figure 5.5

We cannot answer this question. Not because we do not know, but because we have not yet specified the system. Which is the body under consideration ? Do not try to identify forces before you have decided the system. Suppose we concentrate on the state of motion of the boy. We should then concentrate on the forces acting *on* the boy. The forces are listed in the upper half of table (5.1). Instead, if we take the load as the system and discuss the equilibrium of the load,

the list of the forces will be different. These forces appear in the lower half of table (5.1).

Table 5.1

System	Force exerted by	Magnitude of the force	Direction of the force	Nature of the force
Boy	Earth	W	Downward	Gravitational
	Floor	$\propto N$	Upward	Electro-magnetic
	Load	$\propto N_1$	Downward	„
Load	Earth	W'	Downward	Gravitational
	Boy	$\propto N_1$	Upward	Electro-magnetic

One may furnish as much information as one has about the magnitude and direction of the forces. The contact forces may have directions other than normal to the contact surface if the surfaces are rough. We shall discuss more about it under the heading of friction.

Step 3 : Make a Free Body Diagram

Now, represent the system by a point in a separate diagram and draw vectors representing the forces acting on the system with this point as the common origin. The forces may lie along a line, may be distributed in a plane (coplanar) or may be distributed in the space (non-planar). We shall rarely encounter situations dealing with non-planar forces. For coplanar forces the plane of diagram represents the plane of the forces acting on the system. Indicate the magnitudes and directions of the forces in this diagram. This is called a *free body diagram*. The free body diagram for the example discussed above with the boy as the system and with the load as the system are shown in figure (5.6).

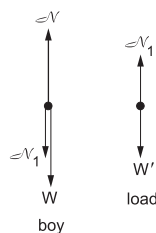


Figure 5.6

Step 4 : Choose Axes and Write Equations

Any three mutually perpendicular directions may be chosen as the X - Y - Z axes. We give below some suggestions for choosing the axes to solve problems.

If the forces are coplanar, only two axes, say X and Y , taken in the plane of forces are needed. Choose the X -axis along the direction in which the system is known to have or is likely to have the acceleration. A direction perpendicular to it may be chosen as the Y -axis. If the system is in equilibrium, any mutually perpendicular directions in the plane of the diagram may be chosen as the axes. Write the components of all the forces along the X -axis and equate their sum to the product of the mass of the system and its acceleration. This gives you one equation. Write the components of the forces along the Y -axis and equate the sum to zero. This gives you another equation. If the forces are collinear, this second equation is not needed.

If necessary you can go to step 1, choose another object as the system, repeat steps 2, 3 and 4 to get more equations. These are called equations of motion. Use mathematical techniques to get the unknown quantities out of these equations. This completes the algorithm.

The magnitudes of acceleration of different objects in a given situation are often related through kinematics. This should be properly foreseen and used together with the equations of motion. For example in figure (5.4) the accelerations of C and E have same magnitudes. Equations of motion for C and for E should use the same variable a for acceleration.

Example 5.2

A block of mass M is pulled on a smooth horizontal table by a string making an angle θ with the horizontal as shown in figure (5.7). If the acceleration of the block is a , find the force applied by the string and by the table on the block.

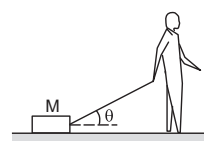


Figure 5.7

Solution : Let us consider the block as the system.

The forces on the block are

- pull of the earth, Mg , vertically downward,
- contact force by the table, $\propto N$, vertically upward,
- pull of the string, T , along the string.

The free body diagram for the block is shown in figure (5.8).

The acceleration of the block is horizontal and towards the right. Take this direction as the X -axis and vertically upward direction as the Y -axis. We have,

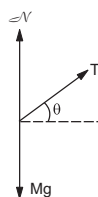


Figure 5.8

component of Mg along the X -axis = 0
 component of N along the X -axis = 0
 component of T along the X -axis = $T \cos \theta$.

Hence the total force along the X -axis = $T \cos \theta$.

Using Newton's law, $T \cos \theta = Ma$ (i)

Component of Mg along the Y -axis = $-Mg$
 component of N along the Y -axis = N
 component of T along the Y -axis = $T \sin \theta$.
 Total force along the Y -axis = $N + T \sin \theta - Mg$.

Using Newton's law, $N + T \sin \theta - Mg = 0$ (ii)

From equation (i), $T = \frac{Ma}{\cos \theta}$. Putting this in equation (ii)

$$N = Mg - Ma \tan \theta.$$

5.4 NEWTON'S THIRD LAW OF MOTION

Newton's third law has already been introduced in chapter 4. "If a body A exerts a force $\vec{F}$ on another body B , then B exerts a force $-\vec{F}$ on A ."

Thus, the force exerted by A on B and that by B on A are equal in magnitude but opposite in direction. This law connects the forces exerted by two bodies on one another. The forces connected by the third law act on two different bodies and hence will never appear together in the list of forces at step 2 of applying Newton's first or second law.

For example, suppose a table exerts an upward force N on a block placed on it. This force should be accounted if we consider the block as the system. The block pushes the table down with an equal force N . But this force acts on the table and should be considered only if we take the table as the system. Thus, only one of the two forces connected by the third law may appear in the equation of motion depending on the system chosen. The force exerted by the earth on a particle of mass M is Mg downward and therefore, by the particle on the earth is Mg upward. These two forces will not cancel each other. The downward force on the particle will cause acceleration of the particle and that on the earth will cause acceleration (how large?) of the earth.

Newton's third law of motion is not strictly correct when interaction between two bodies separated by a large distance is considered. We come across such deviations when we study electric and magnetic forces.

Working with the Tension in a String

The idea of tension was qualitatively introduced in chapter 4. Suppose a block of mass M is hanging through a string from the ceiling (figure 5.9).

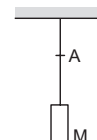


Figure 5.9

Consider a cross-section of the string at A . The cross-section divides the string in two parts, lower part and the upper part. The two parts are in physical contact at the cross-section at A . The lower part of the string will exert an electromagnetic force on the upper part and the upper part will exert an electromagnetic force on the lower part. According to the third law, these two forces will have equal magnitude. The lower part pulls down the upper part with a force T and the upper part pulls up the lower part with equal force T . The common magnitude of the forces exerted by the two parts of the string on each other is called the tension in the string at A . What is the tension in the string at the lower end? The block and the string are in contact at this end and exert electromagnetic forces on each other. The common magnitude of these forces is the tension in the string at the lower end. What is the tension in the string at the upper end? At this end, the string and the ceiling meet. The string pulls the ceiling down and the ceiling pulls the string up. The common magnitude of these forces is the tension in the string at the upper end.

Example 5.3

The mass of the part of the string below A in figure (5.9) is m . Find the tension of the string at the lower end and at A .

Solution : To get the tension at the lower end we need the force exerted by the string on the block.

Take the block as the system. The forces on it are

- (a) pull of the string, T , upward,
- (b) pull of the earth, Mg , downward,

The free body diagram for the block is shown in figure (5.10a). As the acceleration of the block is zero, these forces should add to zero. Hence the tension at the lower end is $T = Mg$.

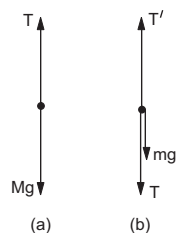


Figure 5.10

To get the tension T at A we need the force exerted by the upper part of the string on the lower part of the string. For this we may write the equation of motion for the lower part of the string. So take the string below A as the system. The forces acting on this part are

- (a) T' , upward, by the upper part of the string
- (b) mg , downward, by the earth
- (c) T , downward, by the block.

Note that in (c) we have written T for the force by the block on the string. We have already used the symbol T for the force by the string on the block. We have used Newton's third law here. The force exerted by the block on the string is equal in magnitude to the force exerted by the string on the block.

The free body diagram for this part is shown in figure (5.10b). As the system under consideration (the lower part of the string) is in equilibrium, Newton's first law gives

$$T' = T + mg$$

But $T = Mg$ hence, $T' = (M + m)g$.

Example 5.4

The block shown in figure (5.11) has a mass M and descends with an acceleration a . The mass of the string below the point A is m . Find the tension of the string at the point A and at the lower end.

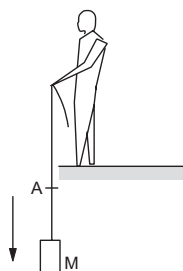


Figure 5.11

Solution : Consider “the block + the part of the string below A” as the system. Let the tension at A be T . The forces acting on this system are

- (a) $(M + m)g$, downward, by the earth
- (b) T , upward, by the upper part of the string.

The first is gravitational and the second is electromagnetic. We do not have to write the force by the string on the block. This electromagnetic force is by one part of the system on the other part. Only the forces acting on the system by the objects other than the system are to be included.

The system is descending with an acceleration a . Taking the downward direction as the X -axis, the total force along the X -axis is $(M + m)g - T$. Using Newton's law

$$(M + m)g - T = (M + m)a.$$

or, $T = (M + m)(g - a)$ (i)

We have omitted the free body diagram. This you can do if you can draw the free body diagram in your mind and write the equations correctly.

To get the tension T' at the lower end we can put $m = 0$ in (i).

Effectively, we take the point A at the lower end. Thus, we get $T' = M(g - a)$.

Suppose the string in **Example 5.3** or **5.4** is very light so that we can neglect the mass of the string. Then $T' = T$. The tension is then the same throughout the string. This result is of general nature. The tension at all the points in a string or a spring is the same provided it is assumed massless and no massive particle or body is connected in between.

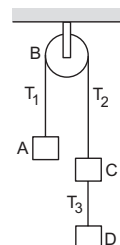


Figure 5.12

If the string in figure (5.12) is light, the tension T_1 of the string is same at all the points between the block A and the pulley B. The tension T_2 is same at all the points between the pulley B and the block C. The tension T_3 is same at all the points between the block C and the block D. The three tensions T_1 , T_2 and T_3 may be different from each other. If the pulley B is also light, then $T_1 = T_2$.

5.5 PSEUDO FORCES

In this section we discuss the techniques of solving the motion of a body with respect to a noninertial frame of reference.

Consider the situation shown in figure (5.13). Suppose the frame of reference S' moves with a

constant acceleration $\vec{a}_0$ with respect to an inertial frame S . The acceleration of a particle P measured with respect to S' is $\vec{a}_{P,S'} = \vec{a}$ and that with respect to S is $\vec{a}_{P,S}$. The acceleration of S' with respect to S is $\vec{a}_{S',S} = \vec{a}_0$.

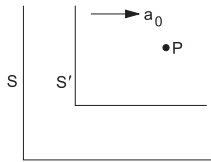


Figure 5.13

Since S' is translating with respect to S we have,

$$\vec{a}_{P,S'} = \vec{a}_{P,S} + \vec{a}_{S',S} = \vec{a}_{P,S} - \vec{a}_{S',S}$$

$$\text{or, } \vec{a} = \vec{a}_{P,S} - \vec{a}_0$$

$$\text{or, } m \vec{a} = m \vec{a}_{P,S} - m \vec{a}_0$$

where m is the mass of the particle P . Since S is an inertial frame $m \vec{a}_{P,S}$ is equal to the sum of all the forces acting on P . Writing this sum as $\vec{F}$, we get

$$m \vec{a} = \vec{F} - m \vec{a}_0$$

$$\text{or, } \vec{a} = \frac{\vec{F} - m \vec{a}_0}{m} \quad \dots (5.3)$$

This equation relates the acceleration of the particle and the forces acting on it. Compare it with equation (5.2) which relates the acceleration and the force when the acceleration is measured with respect to an inertial frame. The acceleration of the frame (with respect to an inertial frame) comes into the equation of a particle. Newton's second law $\vec{a} = \vec{F}/m$ is not valid in such a noninertial frame. An extra term $-m \vec{a}_0$ has to be added to the sum of all the forces acting on the particle before writing the equation $\vec{a} = \vec{F}/m$. Note that in this extra term, m is the mass of the particle under consideration and $\vec{a}_0$ is the acceleration of the working frame of reference with respect to some inertial frame.

However, we people spend most of our lifetime on the earth which is an (approximate) inertial frame. We are so familiar with the Newton's laws that we would still like to use the terminology of Newton's laws even when we use a noninertial frame. This can be done if we agree to call $(-m \vec{a}_0)$ a *force* acting on the particle. Then while preparing the list of the forces acting on the particle P , we include all the (real) forces acting on P by all other objects and also include an imaginary

force $-m \vec{a}_0$. Applying Newton's second law will then lead to equation (5.3). Such correction terms $-m \vec{a}_0$ in the list of forces are called *pseudo forces*. This so-called force is to be included in the list only because we are discussing the motion from a noninertial frame and still want to use Newton's second law as "total force equals mass times acceleration". If we work from an inertial frame, the acceleration $\vec{a}_0$ of the frame is zero and no pseudo force is needed. The pseudo forces are also called *inertial forces* although their need arises because of the use of noninertial frames.

Example 5.5

A pendulum is hanging from the ceiling of a car having an acceleration a_0 with respect to the road. Find the angle made by the string with the vertical.

Solution : The situation is shown in figure (5.14a). Suppose the mass of the bob is m and the string makes an angle θ with the vertical. We shall work from the car frame. This frame is noninertial as it has an acceleration $\vec{a}_0$ with respect to an inertial frame (the road). Hence, if we use Newton's second law we shall have to include a pseudo force.

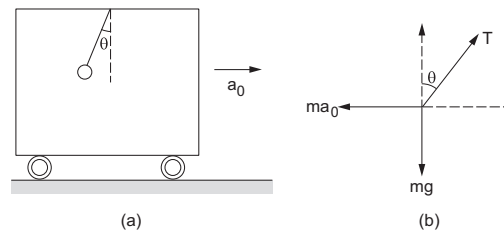


Figure 5.14

Take the bob as the system.

The forces are :

- (a) T along the string, by the string
- (b) mg downward, by the earth
- (c) ma_0 towards left (pseudo force).

The free body diagram is shown in figure (5.14b). As the bob is at rest (remember we are discussing the motion with respect to the car) the force in (a), (b) and (c) should add to zero. Take X -axis along the forward horizontal direction and Y -axis along the upward vertical direction. The components of the forces along the X -axis give

$$T \sin \theta - m a_0 = 0 \quad \text{or, } T \sin \theta = m a_0 \quad \dots (i)$$

and the components along the Y -axis give

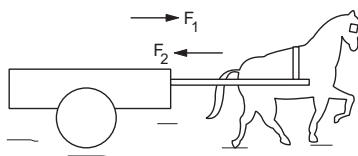
$$T \cos \theta - mg = 0 \quad \text{or, } T \cos \theta = mg. \quad \dots (ii)$$

Dividing (i) by (ii) $\tan \theta = a_0 / g$.

Thus, the string makes an angle $\tan^{-1}(a_0 / g)$ with the vertical.

5.6 THE HORSE AND THE CART

A good example which illustrates the ideas discussed in this chapter is the motion of a cart pulled by a horse. Suppose the cart is at rest when the driver whips the horse. The horse pulls the cart and the cart accelerates forward. The question posed is as follows. The horse pulls the cart by a force F_1 in the forward direction. From the third law of motion the cart pulls the horse by an equal force $F_2 = F_1$ in the backward direction. The sum of these forces is, therefore, zero (figure 5.15). Why should then the cart accelerate forward?



F_1 : Force on the cart by the horse
 F_2 : Force on the horse by the cart
 $F_1 = F_2 = F$

Figure 5.15

Try to locate the mistake in the argument. According to our scheme, we should first decide the system. We can take the horse as the system or the cart as the system or the cart and the horse taken together as the system. Suppose you take the cart as the system. Then the forces on the cart should be listed and the forces on the horse should not enter the discussion. The force on the cart is F_1 in the forward direction and the acceleration of the cart is also in the forward direction. How much is this acceleration? Take the mass of the cart to be M_C . Is the acceleration of the cart $a = F_1/M_C$ in forward direction? Think carefully. We shall return to this question.

Let us now try to understand the motion of the horse. This time we have to consider the forces on the horse. The forward force F_1 by the horse acts on the cart and it should not be taken into account when we discuss the motion of the horse. The force on the horse by the cart is F_2 in the backward direction. Why does the horse go in the forward direction when whipped? The horse exerts a force on the cart in the forward direction and hence the cart is accelerated forward. But the cart exerts an equal force on the horse in the backward direction. Why is the horse not accelerated in backward direction? (Imagine this situation. If the cart is accelerated forward and the horse backward, the horse will sit on the cart kicking out the driver and the passengers.) Where are we wrong? We have not considered *all* the forces acting on the horse. The

road pushes the horse by a force P which has a forward component. This force acts on the horse and we must add this force when we discuss the motion of the horse. The horse accelerates forward if the forward component f of the force P exceeds F_2 (Figure 5.16). The acceleration of the horse is $(f - F_2)/M_h$. We should make sure that *all* the forces acting on the system are added. Note that the force of gravity acting on the horse has no forward component.

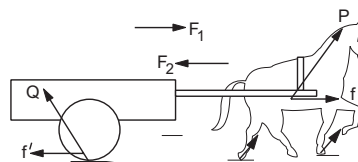


Figure 5.16

Going back to the previous paragraph the acceleration of the cart may not be F_1/M_C . The road exerts a force Q on the cart which may have a backward component f' . The total force on the cart is $F_1 - f'$. The acceleration of the cart is then $a = \frac{F_1 - f'}{M_C}$ in the forward direction.

The forces f and f' are self adjustable and they so adjust their values that $\frac{F_1 - f'}{M_C} = \frac{f - F_2}{M_h}$. The acceleration of the horse and that of the cart are equal in magnitude and direction and hence they move together.

So, once again we remind you that only the forces on the system are to be considered to discuss the motion of the system and all the forces acting on the system are to be considered. Only then apply $\vec{F} = m\vec{a}$.

5.7 INERTIA

A particle is accelerated (in an inertial frame) if and only if a resultant force acts on it. Loosely speaking, the particle does not change its state of rest or of uniform motion along a straight line unless it is forced to do this. This unwillingness of a particle to change its state of rest or of uniform motion along a straight line is called as *inertia*. We can understand the property of inertia in more precise terms as follows. If equal forces are applied on two particles, in general, the acceleration of the particles will be different. The property of a particle to allow a smaller acceleration is called *inertia*. It is clear that larger the mass of the particle, smaller will be the acceleration and hence larger will be the inertia.

CHAPTER 6

FRICTION

6.1 FRICTION AS THE COMPONENT OF CONTACT FORCE

When two bodies are kept in contact, electromagnetic forces act between the charged particles at the surfaces of the bodies. As a result, each body exerts a contact force on the other. The magnitudes of the contact forces acting on the two bodies are equal but their directions are opposite and hence the contact forces obey Newton's third law.

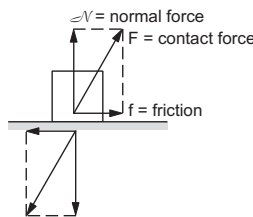


Figure 6.1

The direction of the contact force acting on a particular body is not necessarily perpendicular to the contact surface. We can resolve this contact force into two components, one perpendicular to the contact surface and the other parallel to it (Figure 6.1). The perpendicular component is called the *normal contact force* or *normal force* and the parallel component is called *friction*.

Example 6.1

A body of mass 400 g slides on a rough horizontal surface. If the frictional force is 3.0 N, find (a) the angle made by the contact force on the body with the vertical and (b) the magnitude of the contact force. Take $g = 10 \text{ m/s}^2$.

Solution : Let the contact force on the block by the surface be F which makes an angle θ with the vertical (figure 6.2).

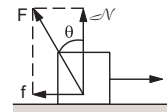


Figure 6.2

The component of F perpendicular to the contact surface is the normal force N and the component of F parallel to the surface is the friction f . As the surface is horizontal, N is vertically upward. For vertical equilibrium,

$$N = Mg = (0.400 \text{ kg}) (10 \text{ m/s}^2) = 4.0 \text{ N}.$$

The frictional force is $f = 3.0 \text{ N}$.

$$(a) \quad \tan \theta = \frac{f}{N} = \frac{3}{4}$$

$$\text{or,} \quad \theta = \tan^{-1} (3/4) = 37^\circ.$$

(b) The magnitude of the contact force is

$$\begin{aligned} F &= \sqrt{N^2 + f^2} \\ &= \sqrt{(4.0 \text{ N})^2 + (3.0 \text{ N})^2} = 5.0 \text{ N}. \end{aligned}$$

Friction can operate between a given pair of solids, between a solid and a fluid or between a pair of fluids. Frictional force exerted by fluids is called *viscous force* and we shall study it in a later chapter. Here we shall study about the frictional forces operating between a pair of solid surfaces.

When two solid bodies slip over each other, the force of friction is called *kinetic friction*. When two bodies do not slip on each other, the force of friction is called *static friction*.

It is difficult to work out a reliable theory of friction starting from the electromagnetic interaction between the particles at the surface. However, a wide range of observations can be summarized in a small number of *laws of friction* which we shall discuss.

6.2 KINETIC FRICTION

When two bodies in contact move with respect to each other, rubbing the surfaces in contact, the friction between them is called kinetic friction. The directions of the frictional forces are such that the relative slipping is opposed by the friction.

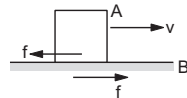


Figure 6.3

Suppose a body A placed in contact with B is moved with respect to it as shown in figure (6.3). The force of friction acting on A due to B will be opposite to the velocity of A with respect to B. In figure (6.3) this force is shown towards left. The force of friction on B due to A is opposite to the velocity of B with respect to A. In figure (6.3) this force is shown towards right. The force of kinetic friction opposes the relative motion. We can formulate the rules for finding the direction and magnitude of kinetic friction as follows :

(a) Direction of Kinetic Friction

The kinetic friction on a body A slipping against another body B is opposite to the velocity of A with respect to B.

It should be carefully noted that the velocity coming into picture is with respect to the body applying the force of friction.

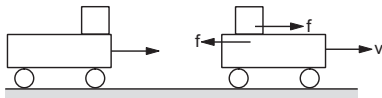


Figure 6.4

As another example, suppose we have a long box having wheels and moving on a horizontal road (figure 6.4). A small block is placed on the box which slips on the box to fall from the rear end. As seen from the road, both the box and the block are moving towards right, of course the velocity of the block is smaller than that of the box. What is the direction of the kinetic friction acting on the block due to the box ? The velocity of the block as seen from the box is towards left. Thus, the friction on the block is towards right. The friction acting on the box due to the block is towards left.

(b) Magnitude of the Kinetic Friction

The magnitude of the kinetic friction is proportional to the normal force acting between the two bodies. We can write

$$f_k = \mu_k \mathcal{N} \quad \dots (6.1)$$

where $\mathcal{N}$ is the normal force. The proportionality constant μ_k is called the *coefficient of kinetic friction* and its value depends on the nature of the two surfaces in contact. If the surfaces are smooth μ_k will be small, if the surfaces are rough μ_k will be large. It also depends on the materials of the two bodies in contact.

According to equation (6.1) the coefficient of kinetic friction does not depend on the speed of the sliding bodies. Once the bodies slip on each other the frictional force is $\mu_k \mathcal{N}$, whatever be the speed. This is approximately true for relative speeds not too large (say for speeds < 10 m/s).

We also see from equation (6.1) that as long as the normal force $\mathcal{N}$ is same, the frictional force is independent of the area of the surface in contact. For example, if a rectangular slab is slid over a table, the frictional force is same whether the slab lies flat on the table or it stands on its face of smaller area (figure 6.5)

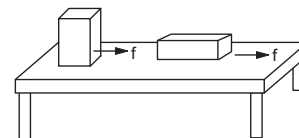


Figure 6.5

Example 6.2

A heavy box of mass 20 kg is pulled on a horizontal surface by applying a horizontal force. If the coefficient of kinetic friction between the box and the horizontal surface is 0.25, find the force of friction exerted by the horizontal surface on the box.

Solution : The situation is shown in figure (6.6). In the vertical direction there is no acceleration, so

$$\mathcal{N} = Mg.$$

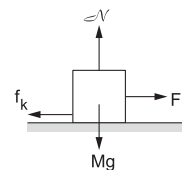


Figure 6.6

As the box slides on the horizontal surface, the surface exerts kinetic friction on the box. The magnitude of the kinetic friction is

$$\begin{aligned} f_k &= \mu_k \mathcal{N} = \mu_k Mg \\ &= 0.25 \times (20 \text{ kg}) \times (9.8 \text{ m/s}^2) = 49 \text{ N.} \end{aligned}$$

This force acts in the direction opposite to the pull.

6.3 STATIC FRICTION

Frictional forces can also act between two bodies which are in contact but are not sliding with respect to each other. The friction in such cases is called *static friction*. For example, suppose several labourers are trying to push a heavy almirah on the floor to take it out of a room (figure 6.7).

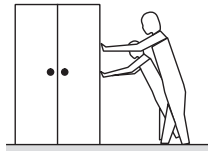


Figure 6.7

The almirah is heavy and even the most sincere effort by them is not able to slide it on the floor even by a millimeter. As the almirah is at rest the resultant force on the almirah should be zero. Thus, something is exerting a force on the almirah in the opposite direction. In this case, it is the floor which exerts a frictional force on the almirah. The labourers push the almirah towards left in figure (6.7) and the floor exerts a frictional force on the almirah towards right. This is an example of static friction.

How strong is this frictional force? Suppose the almirah is pushed with a small force in the beginning and the force is gradually increased. It does not slide until the force applied is greater than a minimum value say F . The force of static friction is equal and opposite to the force exerted by the labourers as long as the almirah is at rest. This means that the magnitude of static friction adjusts its value according to the applied force. As the applied force increases, the frictional force also increases. The static friction is thus, self adjustable. It adjusts its magnitude (and direction) in such a way that together with other forces applied on the body, it maintains 'relative rest' between the two surfaces. However, the frictional force cannot go beyond a maximum. When the applied force exceeds this maximum, friction fails to increase its value and slipping starts. The maximum static friction that a body can exert on the other body in contact with it, is called *limiting friction*. This limiting friction is proportional to the normal contact force between the two bodies. We can write

$$f_{\max} = \mu_s \mathcal{N} \quad \dots (6.2)$$

where $f_{\max}$ is the maximum possible force of static friction and $\mathcal{N}$ is the normal force. The constant of proportionality is called the *coefficient of static friction* and its value again depends on the material and roughness of the two surfaces in contact. In general, μ_s is slightly greater than μ_k . As long as the normal

force is constant, the maximum possible friction does not depend on the area of the surfaces in contact.

Once again we emphasise that $\mu_s \mathcal{N}$ is the **maximum** possible force of static friction that **can** act between the bodies. The actual force of static friction may be smaller than $\mu_s \mathcal{N}$ and its value depends on other forces acting on the body. The magnitude of frictional force is equal to that required to keep the body at relative rest. Thus,

$$f_s \leq f_{\max} = \mu_s \mathcal{N}. \quad \dots (6.3)$$

Example 6.3

A boy (30 kg) sitting on his horse whips it. The horse speeds up at an average acceleration of 2.0 m/s^2 . (a) If the boy does not slide back, what is the force of friction exerted by the horse on the boy? (b) If the boy slides back during the acceleration, what can be said about the coefficient of static friction between the horse and the boy. Take $g = 10 \text{ m/s}^2$.

Solution : (a) The forces acting on the boy are

- (i) the weight Mg .
- (ii) the normal contact force $\mathcal{N}$ and
- (iii) the static friction f_s .

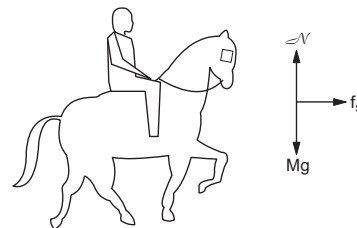


Figure 6.8

As the boy does not slide back, its acceleration a is equal to the acceleration of the horse. As friction is the only horizontal force, it must act along the acceleration and its magnitude is given by Newton's second law

$$f_s = Ma = (30 \text{ kg}) (2.0 \text{ m/s}^2) = 60 \text{ N}.$$

(b) If the boy slides back, the horse could not exert a friction of 60 N on the boy. The maximum force of static friction that the horse may exert on the boy is

$$\begin{aligned} f_s &= \mu_s \mathcal{N} = \mu_s Mg \\ &= \mu_s (30 \text{ kg}) (10 \text{ m/s}^2) = \mu_s 300 \text{ N} \end{aligned}$$

where μ_s is the coefficient of static friction. Thus,

$$\mu_s (300 \text{ N}) < 60 \text{ N}$$

$$\text{or,} \quad \mu_s < \frac{60}{300} = 0.20.$$

Finding the Direction of Static Friction

The direction of static friction on a body is such that the total force acting on it keeps it at rest with respect to the body in contact. Newton's first or second law can often be used to find the direction of static friction. Figure (6.9) shows a block A placed on another block B which is placed on a horizontal table.



Figure 6.9

The block B is pulled by a force F towards right. Suppose the force is small and the blocks do not move. Let us focus our attention on the upper block. The upper block is at rest with respect to the ground which is assumed to be inertial. Thus, the resultant force on the upper block is zero (Newton's first law). As no other external force acts on the upper block the friction acting on the upper block due to the lower block, must be zero. If the force F is increased, the two blocks move together towards right, with some acceleration. As the upper block accelerates towards right the resultant force on it must be towards right. As friction is the only horizontal force on the upper block it must be towards right.

Notice that it is the friction on the upper block which accelerates it towards right. It is a general misconception that friction always opposes the motion. It is not really true. In many cases friction causes the motion. A vehicle accelerates on the road only because the frictional force on the vehicle due to the road drives it. It is not possible to accelerate a vehicle on a frictionless road. Friction opposes the relative motion between the bodies in contact.

Another way to find the direction of static friction is as follows. For a moment consider the surfaces to be frictionless. In absence of friction the bodies will start slipping against each other. One should then find the direction of friction as opposite to the velocity with respect to the body applying the friction.

6.4 LAWS OF FRICTION

We can summarise the laws of friction between two bodies in contact as follows :

(1) If the bodies slip over each other, the force of friction is given by

$$f_k = \mu_k \mathcal{N}$$

where $\mathcal{N}$ is the normal contact force and μ_k is the coefficient of kinetic friction between the surfaces.

(2) The direction of kinetic friction on a body is opposite to the velocity of this body with respect to the body applying the force of friction.

(3) If the bodies do not slip over each other, the force of friction is given by

$$f_s \leq \mu_s \mathcal{N}$$

where μ_s is the coefficient of static friction between the bodies and $\mathcal{N}$ is the normal force between them. The direction and magnitude of static friction are such that the condition of no slipping between the bodies is ensured.

(4) The frictional force f_k or f_s does not depend on the area of contact as long as the normal force $\mathcal{N}$ is same.

Table (6.1) gives a rough estimate of the values of coefficient of static friction between certain pairs of materials. The actual value depends on the degree of smoothness and other environmental factors. For example, wood may be prepared at various degrees of smoothness and the friction coefficient will vary.

Table 6.1 : The Friction Coefficients

Material	μ_s	Material	μ_s
Steel and steel	0.58	Copper and copper	1.60
Steel and brass	0.35	Teflon and teflon	0.04
Glass and glass	1.00	Rubber tyre on dry concrete road	1.0
Wood and wood	0.35	Rubber tyre on wet concrete road	0.7
Wood and metal	0.40		
Ice and ice	0.10		

Dust, impurities, surface oxidation etc. have a great role in determining the friction coefficient. Suppose we take two blocks of pure copper, clean them carefully to remove any oxide or dust layer at the surfaces, heat them to push out any dissolved gases and keep them in contact with each other in an evacuated chamber at a very low pressure of air. The blocks stick to each other and a large force is needed to slide one over the other. The friction coefficient as defined above, becomes much larger than one. If a small amount of air is allowed to go into the chamber so that some oxidation takes place at the surface, the friction coefficient reduces to usual values.

6.5 UNDERSTANDING FRICTION AT ATOMIC LEVEL

It has already been pointed out that friction appears because of the interaction between the charged particles of the two bodies near the surfaces of contact. Any macroscopic object like a steel plate or a wood piece has irregular surface at atomic scale. A polished steel surface may look plane to naked eyes but if seen

under a powerful microscope, its surface is found to be quite irregular. Figure (6.10) shows qualitatively how an apparently plane surface may be at the atomic scale.



Figure 6.10

When two bodies are kept one over the other, the real area in contact is much smaller than the total surface area of the bodies (figure 6.11) The distance between the particles of the two bodies becomes very small at these actual points of contact and the molecular forces start operating across the surface. Molecular bonds are formed at these contact points. When one of the two bodies is pulled over the other, these bonds are broken, the materials under the bond is deformed and new bonds are formed. The local deformation of the bodies send vibration waves into the bodies. These vibrations finally damp out and the energy appears as the increased random motion of the particles of the bodies. The bodies thus, become heated. A force is, therefore, needed to start the motion or to maintain the motion.

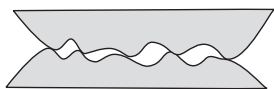


Figure 6.11

6.6 A LABORATORY METHOD TO MEASURE FRICTION COEFFICIENT

(a) Horizontal Table Method

Figure (6.12) shows the apparatus. A wooden plank A is fixed on a wooden frame kept on a table. A frictionless pulley is fixed to one end of the plank. A block B is kept on the plank and is attached to a hanger H by a string which passes over the pulley.

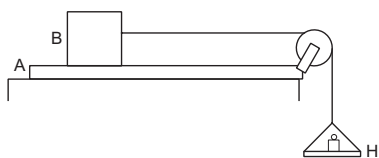


Figure 6.12

The plank is kept in a horizontal position. The friction coefficient between the block B and the plank A can be obtained by using this apparatus.

The weights of the block B and the hanger H are measured. Standard weights are kept on the hanger. The weights are gradually increased and the minimum weight needed to just slide the block is noted.

Suppose the weight of the block is W_1 and the weight of the hanger together with the standard weights is W_2 when the block just starts to slide. The tension in the string is W_2 and same is the force of friction on the block by the plank. Thus, the maximum force of static friction on the block is $f_{\max} = W_2$. The normal force on the block by the plank is equal to the weight of the block itself as the block is in vertical equilibrium. Thus, the normal force is $\mathcal{N} = W_1$.

The coefficient of static friction is

$$\mu_s = \frac{f_{\max}}{\mathcal{N}} = \frac{W_2}{W_1}.$$

To obtain the coefficient of kinetic friction, the weight on the hanger is slightly reduced and the block is gently pushed with a finger to move it on the plank. The weight on the hanger is so adjusted that once pushed, the block continues to move on the plank with uniform speed. In this case, the tension in the string equals the force of kinetic friction. As the hanger also moves with uniform velocity, the tension equals the weight of the hanger plus the standard weights kept in it. For vertical equilibrium of the block, the normal force on the block equals the weight of the block. Thus, if W_1 is the weight of the block and W_2' is the weight of the hanger plus the standard weights, the coefficient of kinetic friction is

$$\mu_k = \frac{f_k}{\mathcal{N}} = \frac{W_2'}{W_1}.$$

One can put certain standard weights on the block to increase the normal force and repeat the experiment. It can be verified that the force of friction also increases and $f_k / \mathcal{N}$ comes out to be the same as it should be because the nature of the surfaces is same. If the block is kept on the plank on some other face, the area of contact is changed. It can be verified by repeating the above experiment that the force of friction does not depend on the area of contact for a given value of normal contact force.

(b) Inclined Table Method

In this method no pulley is needed. A wooden plank A is fixed on a wooden frame. One end of the plank is fixed to the frame on a hinge and the other end can be moved vertically and can be fixed at the desired position. Thus, the plank can be fixed in an inclined position and the angle of incline can be adjusted. A schematic diagram is shown in figure (6.13).

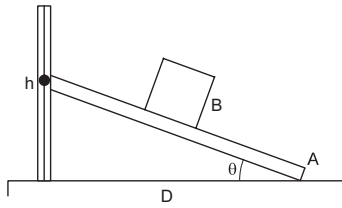


Figure 6.13

Block B is placed on the incline and the angle of the incline is gradually increased. The angle of the incline is so adjusted that the block just starts to slide. The height h and the horizontal distance D between the two ends of the plank are measured. The angle of incline θ satisfies

$$\tan\theta = h/D.$$

Let m be the mass of the block. The forces on the block in case of limiting equilibrium are (figure 6.14)

- (i) weight of the block mg ,
- (ii) the normal contact force $\mathcal{N}$, and
- (iii) the force of static friction f_s .

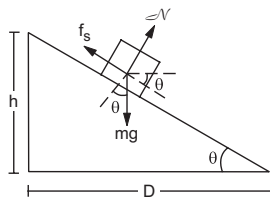


Figure 6.14

Taking components along the incline and applying Newton's first law,

$$f_s = mg \sin\theta.$$

Taking components along the normal to the incline,

$$\mathcal{N} = mg \cos\theta.$$

Thus, the coefficient of static friction between the block and the plank is

$$\mu_s = \frac{f_s}{\mathcal{N}} = \frac{mg \sin\theta}{mg \cos\theta} = \tan\theta = \frac{h}{D}.$$

To obtain the kinetic friction, the inclination is reduced slightly and the block is made to move on the plank by gently pushing it with a finger. The inclination is so adjusted that once started, the block continues with uniform velocity on the plank. The height h' and the distance D' are noted. An identical analysis shows that the force of kinetic friction is

$$f_k = mg \sin\theta$$

and the normal contact force is

$$\mathcal{N} = mg \cos\theta$$

so that the coefficient of kinetic friction between the block and the plank is

$$\mu_k = \frac{f_k}{\mathcal{N}} = \tan\theta = h'/D'.$$

Example 6.4

A wooden block is kept on a polished wooden plank and the inclination of the plank is gradually increased. It is found that the block starts slipping when the plank makes an angle of 18° with the horizontal. However, once started the block can continue with uniform speed if the inclination is reduced to 15° . Find the coefficients of static and kinetic friction between the block and the plank.

Solution : The coefficient of static friction is

$$\mu_s = \tan 18^\circ$$

and the coefficient of kinetic friction is

$$\mu_k = \tan 15^\circ$$

Rolling Friction

It is quite difficult to pull a heavy iron box on a rough floor. However, if the box is provided with four wheels, also made of iron, it becomes easier to move the box on the same floor.

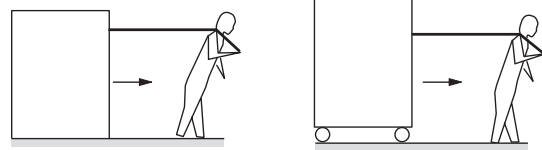


Figure 6.15

The wheel does not slide on the floor rather it rolls on the floor. The surfaces at contact do not rub each other. The velocity of the point of contact of the wheel with respect to the floor remains zero all the time although the centre of the wheel moves forward. The friction in the case of rolling is quite small as compared to kinetic friction. Quite often the rolling friction is negligible in comparison to the static or kinetic friction which may be present simultaneously. To reduce the wear and tear and energy loss against friction, small steel balls are kept between the rotating parts of machines which are known as ball bearings (figure 6.16).

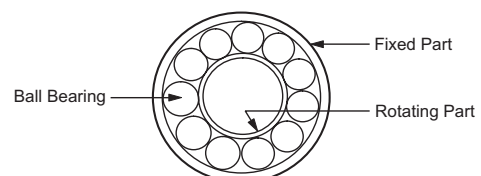


Figure 6.16